scientific reports

OPEN

Multi-modal and multi-agent reinforcement learning framework for urban traffic flow prediction and signal control optimization

Ruokun Wang[1]✉, Ju Zhang[1], Xikui Wang[1], Haoyang He[2] & Yuelin Zou[3]

The rapid urbanization of cities has exacerbated traffic congestion, resulting in significant environmental impacts, including elevated greenhouse gas emissions and deteriorating air quality. Traffic management systems, while effective in many contexts, often fail to consider the ecological and dynamic complexities of modern urban environments. This paper introduces MM-STMAP, a framework for urban traffic management that integrates multi modal perception with deep reinforcement learning. The approach utilizes a spatio temporal graph convolutional network to model intricate traffic patterns across diverse urban environments, while incorporating real-time environmental data, including meteorological factors, to address the ecological limitations of traditional traffic systems. A linear attention mechanism is employed to optimize computational efficiency in processing large-scale, dynamic traffic data, thereby enhancing both operational performance and energy consumption. The multi agent reinforcement learning structure governs the coordination of traffic signals across intersections, achieving a dual optimization of reduced vehicular delays and minimized emissions. Empirical evaluations on major metropolitan datasets demonstrate that MM-STMAP outperforms existing traffic management methods and significantly enhances traffic flow efficiency. The model's ability to integrate heterogeneous data streams spanning traffic sensors and environmental reports enables a comprehensive and adaptive approach to urban mobility, supporting the development of sustainable smart city infrastructure.

Keywords Urban traffic management, Multi-modal learning, Spatiotemporal graph convolutional network, Multi-agent reinforcement learning, Traffic flow prediction

In the context of accelerating urbanization, traffic issues have become one of the major constraints on urban development. With population growth and the expansion of urban areas, traffic volume continues to rise, leading to severe problems such as traffic congestion, environmental pollution, and energy wastage[1,2]. These issues not only affect the efficiency of residents' daily travel but also place enormous pressure on the economy and social life of cities. To address these challenges, Intelligent Transportation Systems (ITS) have gradually become a core component of modern urban management. ITS can optimize traffic flow, reduce traffic accidents, and improve road efficiency by integrating advanced information technology, big data analysis, and real-time monitoring systems[3,4]. Especially in the fields of traffic flow prediction and signal control optimization, ITS has shown great potential in improving traffic management efficiency, alleviating traffic pressure, and enhancing urban travel sustainability.

A critical gap in current research is the decoupling of traffic flow prediction and signal control optimization: existing methods treat these two core tasks independently, failing to leverage predictive insights to guide adaptive control decisions, which directly limits the efficiency of urban traffic management. Addressing this integrated optimization challenge is essential for alleviating persistent traffic congestion and improving the sustainability of urban mobility–key priorities for modern ITS development. Statistical approaches like Autoregressive Integrated Moving Average (ARIMA) and Vector Autoregression (VAR) remain prevalent in traffic flow forecasting. ARIMA leverages historical traffic patterns, demonstrating reliability for systems with straightforward dynamics[5,6]. VAR's strength lies in examining interdependent relationships across multiple roadway segments through

[1]School of Communication and Signal, Nanjing Vocation Institute of Railway Technology, Nanjing 210031, China. [2]Glarun Technology Co., Ltd., Nanjing 210039, Jiangsu, China. [3]State Grid Xinjiang Information & Telecommunication Company, Urumqi 830000, Xinjiang, China. ✉email: 2014560010@njrts.edu.cn

time-series analysis[7]. These traditional methods reveal significant limitations when applied to real-world traffic scenarios. Nonlinear patterns and complex spatial-temporal interactions often undermine their predictive accuracy[8,9]. Sudden traffic fluctuations particularly challenge these rigid frameworks, as they lack the flexibility to adapt to rapidly changing urban traffic conditions where numerous variables coexist. GCNs represent traffic networks as graphs, with nodes symbolizing road segments and intersections[10–12]. This structure helps analyze spatial relationships across urban traffic systems. LSTM and Transformer-based approaches excel at identifying long-term temporal patterns. They effectively address recurring traffic trends and seasonal fluctuations. Still, computational efficiency remains an issue, particularly for large-scale, dynamic networks. Processing lengthy time-series data and adjusting signal control in real time present ongoing challenges[13,14].

Motivated by this key research gap and the urgent need for efficient integrated traffic management solutions, this work aims to develop a unified framework that synergizes spatiotemporal prediction with multi-agent signal control, while addressing computational inefficiencies in long time-series processing. To address the challenges of simultaneously optimizing traffic flow prediction and signal control in current urban traffic systems, this paper proposes a multimodal perception-driven spatiotemporal modeling and reinforcement learning fusion model, named MM-STMAP (Multi-Modal Spatio-Temporal MAPPO). This model integrates the spatiotemporal graph convolution network to model the spatial structure and dynamic evolution of traffic networks, combined with a weighted linear attention mechanism, which enhances the modeling efficiency and expressiveness when processing long time-series data. Additionally, the model incorporates multiple sources of traffic-related modal information, including traffic flow, weather factors, and holiday events, to improve its ability to comprehensively perceive traffic states in complex environments. In the decision optimization layer, MM-STMAP employs Multi-Agent Proximal Policy Optimization (MAPPO) to model the signal controllers of multiple intersections as autonomous agents. Through joint training and strategy coordination, it generates efficient and dynamic traffic signal control strategies. This comprehensive framework not only improves the accuracy of traffic flow prediction but also significantly optimizes the flexibility and real-time performance of signal timing, providing a more intelligent and scalable solution for traffic management in complex urban environments. Notably, MM-STMAP achieves three levels of technical innovation. Structural innovation lies in constructing a closed loop collaborative architecture of perception modeling decision. Learning mechanism innovation involves designing adaptive attention and constrained multi agent learning. System integration innovation realizes deep fusion of multi modal information and multi agent control.

The main contributions of this paper are threefold:

- We propose a new spatiotemporal graph convolutional network architecture with bidirectional synergy with decision optimization. It not only effectively captures the spatiotemporal dependencies of traffic flow to improve prediction accuracy but also dynamically adjusts spatial temporal kernel parameters based on the reward feedback from multi agent control, solving the feature decision mismatch problem in direct module combinations.
- We introduce a weighted linear attention mechanism with dynamic modal adaptation. It optimizes the handling of long time series data by reducing computational complexity from quadratic to linear, addressing the efficiency bottleneck of traditional self attention mechanisms. It further adaptively adjusts the weight of multi modal information such as weather and holidays based on spatiotemporal context, enhancing the model's responsiveness to dynamic traffic changes.
- We employ multi agent reinforcement learning with spatiotemporal constraint embedding to optimize traffic signal control strategies. By integrating spatial temporal dependency features from the spatiotemporal graph convolutional network into the agent reward function and decision sharing mechanism, it enables signal timings to be adjusted based on both real time local traffic flow and network wide dynamics, addressing the adaptability issues of traditional control methods and the local global decoupling problem in standard multi agent reinforcement learning.

Through experimental validation on multiple urban traffic datasets, the proposed method demonstrates superior performance in both traffic flow prediction and signal control optimization, offering new insights and methods for the future management of urban intelligent transportation systems.

Related work

Research on multi-source traffic information perception and fusion mechanisms

Multi-source perception and modality fusion are key directions for intelligent transportation systems and have attracted widespread attention recently[15,16]. Traditional traffic modeling relies on basic data including flow, speed and density, while heterogeneous factors such as weather, holidays, construction and events are now recognized as critical to traffic dynamics[17]. Fusing such heterogeneous data–featuring modal heterogeneity, temporal dynamics and semantic differences–effectively enhances model adaptability and prediction robustness in complex environments. Multi-modal traffic modeling research focuses on three core aspects: data representation, modality fusion strategies and spatiotemporal joint modeling. Typical data representation methods embed structured and unstructured data into a unified space for cross-modal modeling, and some studies use graph structures or regional features to encode spatial relationships between modalities[18,19]. Recent spatio-temporal graph neural networks like DCRNN[20], PIGAT[21] and ASTGCN[22] improve traffic flow prediction by capturing spatiotemporal dependencies in complex traffic systems.

Fusion mechanisms in the literature typically adopt one of three strategies: early fusion, intermediate fusion, or late fusion. Early fusion directly concatenates modality features, which is simple but struggles with dynamic inconsistencies between modalities[23,24]. Intermediate fusion methods process each modality through independent feature extraction subnetworks, then introduce shared spaces or interaction structures for joint modeling. This approach, especially in heterogeneous traffic scenar ios, provides robust modality expression. Late fusion integrates and optimizes decision or output layers, using techniques like weighted averaging or attention to improve system stability. Linear and efficient attention mechanisms, such as Performer[25,26] and Linear Transformer[27], have been introduced to address the high computational cost of self-attention in large-scale traffic systems, allowing more efficient processing of long time-series data. Attention-based fusion strategies have shown improved adaptability in modality selection and dynamic weighting[28]. These methods adjust the contribution of each modality based on its relevance in the current spatiotemporal context. This approach not only mitigates redundancy but also amplifies key modalities' response in the face of sudden events. For example, during adverse weather or holidays, the model increases the weight of external data, enhancing prediction accuracy[29]. Event-aware GNN methods, such as those incorporating public holidays, further improve prediction by integrating event-driven data into traffic modeling[30,31]. Additionally, multi-modal data shows both complementary characteristics at the feature level and dynamic synergistic features at the temporal level[32–34]. Some studies model the synchronous relationships between modalities using temporal convolutions, sliding window alignment, and cross-modal position encoding to explore the interaction patterns between traffic, weather, and holidays[35–37]. This spatiotemporal joint modeling significantly reduces prediction error propagation and improves the ability to model non-stationary traffic flows, particularly in dynamic traffic environments.

Overall, multi-source traffic perception and fusion mechanisms are advancing from simple information concatenation to dynamic, structured, and attention-driven approaches. This progression allows models to gain richer semantic support, which is foundational for subsequent prediction and decision-making modules. The MM-STMAP model proposed in this paper follows this trend, enhancing the ability to represent traffic conditions in complex urban environments by constructing scalable multi-modal embedding modules and integrating graph structures and attention mechanisms.

The evolution of deep reinforcement learning in urban traffic control

Urban traffic signal control is a key mechanism for optimizing traffic flow, aiming to reduce vehicle delay times, shorten queue lengths, improve intersection throughput, and ensure public transportation priority. Traditionally, rule-based control strategies (such as fixed timing and actuated control) and optimization model-based timing algorithms (such as the max pressure algorithm and mixed-integer programming) have dominated the field[38–40]. However, these methods have limited adaptability to changes in traffic conditions, as they rely on predefined rules and struggle to cope with increasingly dynamic and complex traffic environments.

With the development of deep learning technologies, Deep Reinforcement Learning (DRL) has emerged as an innovative approach for urban traffic signal control. Based on trial-and-error learning, DRL uses deep neural networks to approximate policy or value functions in high-dimensional state spaces, enabling agents to autonomously learn efficient control strategies in unknown environments[41–43]. Within a DRL framework, agents map traffic states (such as traffic flow density, queue lengths, and current signal phases) to actions (signal timing decisions) and optimize system performance by maximizing long-term cumulative rewards. DRL models exhibit strong generalization ability and adaptability[44,45]. In single-agent control research, early works mostly adopted value-based methods, such as Deep Q Networks (DQN), to estimate the future value of each action and independently optimize single-intersection signal control[46,47]. These methods are suitable for simple structures and stable traffic features but face bottlenecks in multi-intersection coordinated control, such as state space explosion and lack of global coordination ability.

To address these issues, policy gradient methods have been gradually introduced. One such method is Proximal Policy Optimization (PPO), based on probabilistic policy, which directly optimizes the policy in an end-to-end manner[48,49]. PPO performs more stably in continuous action spaces and has strong training efficiency and convergence properties. The essence of urban traffic is highly multi-agent, with each intersection's signal light acting as an independent control unit, which spatially depends on neighboring intersections. Consequently, the extension of DRL to multi-agent systems has become a research hotspot. Multi-Agent Reinforcement Learning (MARL) constructs multiple autonomous agents, giving them independent observation, decision-making, and learning capabilities[50–52]. Through information sharing and strategy coordination, these agents collaborate to control multiple intersections. Common multi-agent architectures include Independent Learners, Centralized Training with Distributed Execution (CTDE), and Shared Policy/Shared Network Structure, each balancing training efficiency with communication complexity[53–55]. At the algorithmic level, multi-agent variants of PPO, such as Multi-Agent Proximal Policy Optimization (MAPPO), have received widespread attention. MAPPO enhances the stability of policy updates by clipping the range of policy changes, supports distributed deployment, and adapts to the heterogeneity of intersections in complex road networks[56–58]. MAPPO also has good scalability in large-scale systems, high sample utilization, fast convergence, and is well-suited for high-frequency, real-time traffic control requirements. Recent advancements have been made in multi-agent traffic signal control. Notable methods such as PressLight, CoLight, MPLight, and MA2C have enhanced the efficiency of multi-agent systems in real-time decision-making for urban traffic management, optimizing signal control across multiple intersections[30,59,60].

Despite the flexibility of deep reinforcement learning, urban traffic systems face challenges like noise, underfitting, and overfitting. To overcome these, methods such as imitation learning, expert initialization, and graph neural networks are used to improve policy learning and generalization[61]. In summary, deep reinforcement learning is evolving from single-point control to multi-agent collaborative control. The MM-STMAP model in this paper coordinates intersection agents through MAPPO, optimizing signal timing, enhancing decision efficiency, and improving system responsiveness, thereby providing intelligent control strategies for dynamic traffic systems.

Methods

Overall framework of the MM-STMAP model

The MM-STMAP model proposed in this paper aims to enhance the collaborative efficiency of urban traffic flow prediction and signal control through multimodal information fusion and deep reinforcement learning optimization. The entire model is divided into three main modules: the Perception Module, the Spatio-Temporal Modeling Module, and the Decision Optimization Module. These modules collaborate with each other to build an efficient, dynamic, and scalable urban traffic management system. MM-STMAP integrates multimodal traffic information (including historical traffic flow, weather conditions, holidays, and special events) as external environmental factors. The model processes and integrates different types of data to understand the traffic environment. This provides input for modeling and decision-making. With multimodal information, the model captures the complexity and dynamic changes in traffic conditions. The spatiotemporal modeling module uses a spatiotemporal graph convolutional network (STGCN) to model the spatial structure and evolution of the transportation network. It learns spatial dependencies between road segments through graph convolution. Time convolution captures the traffic flow trends. A weighted linear attention mechanism is used to improve the handling of long time series data. This mechanism adjusts the weights of various modal information at different times, helping the model learn spatiotemporal dependencies in complex traffic environments. In the decision optimization module, MM-STMAP adopts the Multi Agent Near End Policy Optimization (MAPPO) algorithm for learning and optimizing signal control strategies. Each traffic signal is treated as an independent intelligent agent that adjusts its signal timing based on local traffic conditions and the behavior of other intelligent agents. Through joint training and strategy coordination, intelligent agents can achieve global traffic flow optimization while maintaining local autonomy. During the training process, MAPPO ensures stability and efficiency through strategy pruning and reward mechanisms, thereby ensuring real-time response of signal control strategies in practical applications. The overall framework in Fig. 1 illustrates the interrelationships between the modules and the flow of information. The Perception Module first collects and fuses multimodal data, passing it to the Spatio-Temporal Modeling Module for learning spatio-temporal dependencies. The Decision Optimization Module then generates the optimal signal control strategy based on the learned states and patterns. This process enables MM-STMAP to efficiently and accurately perform traffic flow prediction and signal timing optimization in complex urban traffic environments.

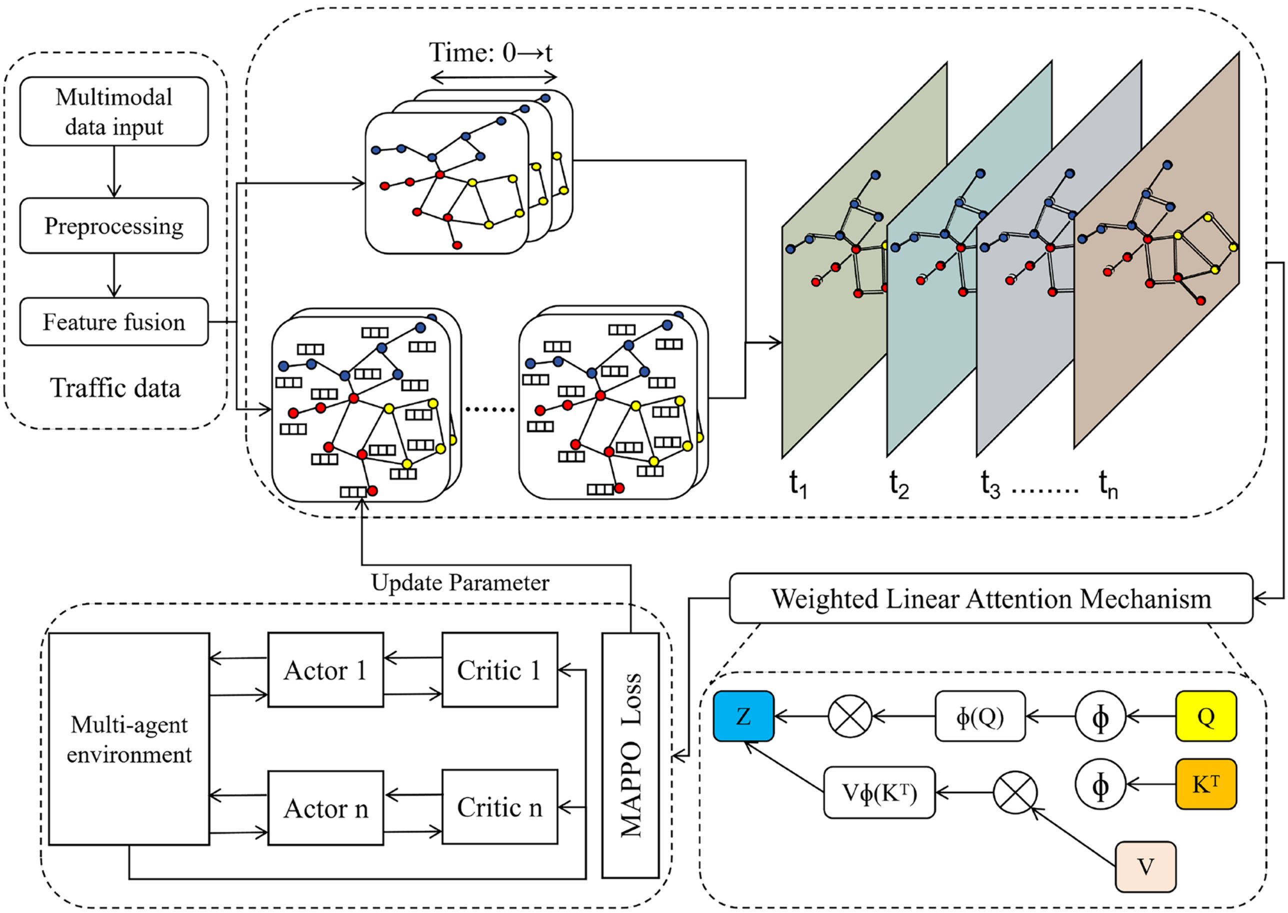


Fig. 1. Overall workflow of the proposed MM-STMAP model.

Spatio-temporal graph convolutional modeling module

The MM-STMAP model's core component is the spatio-temporal graph convolutional modeling module, which captures spatial and temporal dependencies in the traffic network–including long-range temporal correlations that are critical for traffic flow prediction. Moreover, the module is specifically designed to handle non-stationary traffic patterns and sparse data scenarios, which are common in real-world urban traffic systems. The traffic network can be represented as a graph: nodes correspond to traffic segments, and edges denote connections between road segments. Spatio-temporal graph convolutional operations are employed to effectively capture the spatio-temporal characteristics of traffic flow.

For a traffic network graph $G = (V, E)$, where V is the set of nodes (road segments) and E is the set of edges (connections between road segments), each node v has a feature representation $X_v(t) \in \mathbb{R}^C$ at time step t, where C denotes the feature dimension. The adjacency matrix $A \in \mathbb{R}^{|V| \times |V|}$ describes the connection strength between nodes, and we introduce the normalized graph Laplacian $\mathscr{L} = \mathrm{I} - \mathrm{D}^{-1/2} A \mathrm{D}^{-1/2}$ where D is the degree matrix of G to avoid scale distortion in spatial convolution. Notably, the normalized graph Laplacian ensures stable feature propagation even when the adjacency matrix is sparse, laying the foundation for handling sparse data scenarios. The module adopts an alternating structure of spatial graph convolution and temporal convolution to model spatio-temporal dependencies hierarchically. Notably, $X_v(t)$ integrates multi-modal features, including real-time traffic flow, weather conditions, and holiday flags, which are preprocessed via normalization and time alignment before being fed into the module. Specifically, categorical multi-modal variables (e.g., weather types: sunny/rainy/snowy; holiday flags: yes/no) are converted into dense low-dimensional vectors using learnable embedding layers with dimension 8, while continuous variables (e.g., temperature, precipitation intensity) are normalized to the range [0,1] and projected into the same 8-dimensional space via linear transformation. These encoded multi-modal features are then concatenated with the 16-dimensional traffic flow feature vector of the corresponding time step, forming a 24-dimensional combined feature vector. A subsequent feature fusion layer (a two-layer MLP with ReLU activation) maps the combined vector to the target dimension C matching the graph network input, ensuring effective fusion of heterogeneous data types. These multi-modal features provide complementary information to mitigate the impact of traffic non-stationarity caused by external factors.

Firstly, graph convolution operations are used to capture the spatial dependencies between different traffic segments in the network. Given a traffic network graph $G = (V, E)$, where V is the set of nodes (road segments) and E is the set of edges (connections between road segments), each node v has a feature representation $h_v(t)$, where t is the time step. The graph convolution operation propagates node information through the adjacency matrix A. Specifically, the updated feature for node v at time $t+1$ can be written as:

$$h_v^{\text{spatial}}(t+1) = \sigma\left(\sum_{j \in \mathscr{N}(v)} a_{ij} W h_j(t)\right) \tag{1}$$

(Spatial convolution output of node v at time $t+1$)

where $h_v(t)$ represents the feature of node v at time t, a_{ij} is the element of the adjacency matrix representing the connection strength between node i and node j, W is the weight matrix of the graph convolution layer, and σ is the activation function (such as ReLU). The graph convolution operation aggregates the features of neighboring nodes, learning the spatial relationships between nodes. This formula captures the spatial dependencies between traffic segments. This operation generalizes to the batch-wise form $\mathrm{H}_{\text{spatial}}(t) = \sigma\left(\mathscr{L}\mathrm{H}(t)\mathrm{W}\right)$, where $\mathrm{H}_{\text{spatial}}(t)$ is the batch-wise spatial convolution feature matrix, ensuring consistent spatial feature propagation. For sparse data nodes, it leverages adjacent node features to fill gaps, enhancing robustness.

Next, to model the temporal characteristics of traffic flow–including long-range temporal dependencies–we introduce temporal convolutions. Assuming that the historical data $h_v(t-k)$ of traffic flow affects the update of node states, temporal convolution can be written as:

$$h_v^{\text{temporal}}(t+1) = \sigma\left(\sum_{k=1}^{T} b_k h_v(t-k)\right) \tag{2}$$

(Temporal convolution output of node v at time $t+1$)

where b_k represents the weights of the temporal convolution kernel, and $h_v(t-k)$ represents the feature of node v at time $t-k$. In this way, the model can capture the temporal patterns in the traffic flow over time. Temporal convolution utilizes past traffic data to capture the trends and periodic changes in traffic flow, making the model more adaptable in dynamic traffic environments.

We employ 1D causal convolution with kernel size K to model sequential dependencies without information leakage, with the batch-wise formulation $\mathrm{H}_{\text{temp}}(t) = \sigma\left(\mathrm{Conv1d}\left(\mathrm{H}_{\text{spatial}}(t-K+1:t)\right)\right)$, where the convolution operation is applied along the time dimension to capture local temporal dynamics in traffic flow evolution. To extend the receptive field for long-range temporal dependencies, we stack multiple causal convolution layers with dilated rates, enabling the model to capture temporal patterns across distant time steps (e.g., diurnal and weekly traffic fluctuations) without increasing computational complexity. This design helps the model adapt to non-stationary traffic patterns by capturing both short-term fluctuations and long-term periodic trends.

To simultaneously model both spatial and temporal dependencies, we combine graph convolution and temporal convolution. At this point, the updated state of node v at time $t+1$ is influenced by both the spatial neighbors and the historical data from previous time steps, which can be expressed as:

$$h_v(t+1) = \sigma\left(\sum_{j\in\mathcal{N}(v)} a_{ij}Wh_j(t) + \sum_{k=1}^{T} b_k h_v(t-k)\right) \tag{3}$$

This formula simultaneously considers the propagation of spatial information and the temporal information of nodes, thus capturing the spatio-temporal dependencies. By using this spatio-temporal graph convolution approach, the model can accurately capture the spatio-temporal evolution of traffic flow in complex urban traffic networks, including long-range temporal correlations. The module executes spatial convolution and temporal convolution in sequence for each spatio-temporal block, with the overall feature update rule $\mathrm{H}(t+1) = \mathrm{H}_{\mathrm{temp}}(t) + \mathrm{H}(t)$, where residual connections are added to alleviate the gradient vanishing problem in deep spatio-temporal modeling–ensuring stable learning of long-range temporal dependencies in deep layers, which is crucial for adapting to non-stationary traffic dynamics.

To improve the model's ability to handle long time sequences, we introduce a weighting mechanism that dynamically adjusts the influence of historical data on the current node state, further enhancing the capture of long-range temporal dependencies. In this mechanism, the update of node v at time $t+1$ can be written as:

$$h_v(t+1) = \sigma\left(\sum_{j\in\mathcal{N}(v)} a_{ij}Wh_j(t) + \sum_{k=1}^{T} \alpha_k b_k h_v(t-k)\right) \tag{4}$$

where α_k is the dynamically computed coefficient via the weighted mechanism, representing the influence of the past k-th time step on the current node state. The dynamic coefficient α_k is integrated into the temporal convolution kernel weights, resulting in $\tilde{b}_k = \alpha_k \cdot b_k$, which enables adaptive adjustment of temporal dependency strength for traffic flow data. Specifically, it assigns higher weights to critical long-range historical time steps (e.g., traffic data from the same time slot on previous days) and suppresses irrelevant short-term noise, effectively strengthening the model's ability to capture long-range temporal correlations. The weighted mechanism also adaptively assigns weights to multi-modal features based on spatio-temporal context (e.g., increasing weather data weight in rainy conditions), enhancing the model's responsiveness to external factors that affect long-term traffic patterns and mitigate non-stationarity.

This module provides spatio-temporal feature representations for MM-STMAP, significantly enhancing the accuracy of traffic flow prediction and the model's adaptability to dynamic changes in traffic flow–especially the ability to model long-range temporal dependencies, handle non-stationary traffic patterns, and adapt to sparse data scenarios–thereby providing strong support for the subsequent decision optimization module. The final output of this module is high-dimensional spatio-temporal features, which are fed into the subsequent prediction head to generate short-term traffic flow trend inference results, and these features are also directly input into the MAPPO multi-agent control layer for signal timing optimization. These features fully incorporate long-range temporal patterns and are robust to data sparsity, supporting proactive traffic control in complex real-world scenarios.

Weighted linear attention mechanism

The weighted linear attention mechanism is a key module in the MM-STMAP model that enhances the efficiency of modeling long-term dependencies. Traditional self-attention mechanisms have a high computational complexity, particularly when dealing with long time series, where the computation and memory requirements grow quadratically. To address this issue, the weighted linear attention mechanism reduces the computational complexity to linear while maintaining the advantages of self-attention in capturing long-term dependencies, making it particularly suitable for large-scale, dynamically changing traffic data. The structure of the weighted linear attention mechanism, which effectively reduces computational complexity while maintaining the advantages of self-attention, is depicted in the following Fig. 2.

In the standard self-attention mechanism, the "attention" value between each node and other nodes is computed. Let the query (Q), key (K), and value (V) matrices be defined. The attention values are computed by taking the dot product. The traditional method's computation formula is:

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

where Q is the query matrix, K is the key matrix, V is the value matrix, and d_k is the dimension of the key. The softmax operation is used to normalize the attention weights. This formula has a computational complexity of $O(N^2 D)$ (N is the sequence length, D is the feature dimension), where N is the number of nodes, which becomes costly as the number of nodes increases.

To adapt to the inherent spatiotemporal duality and dynamic variability of traffic data, the proposed attention mechanism introduces decoupled learnable weights that explicitly model spatial proximity and temporal dynamics respectively. The spatial weight α_i is initialized based on the traffic graph adjacency matrix to encode inherent road network connections, while the temporal weight β_i adopts a time-decay initializer to prioritize recent traffic states–aligning with real-world traffic evolution patterns from the onset of training. This design embeds domain-specific prior knowledge into the attention computation, enabling adaptive modulation of feature contributions without relying on generic approximation strategies. For non-stationary traffic scenarios,

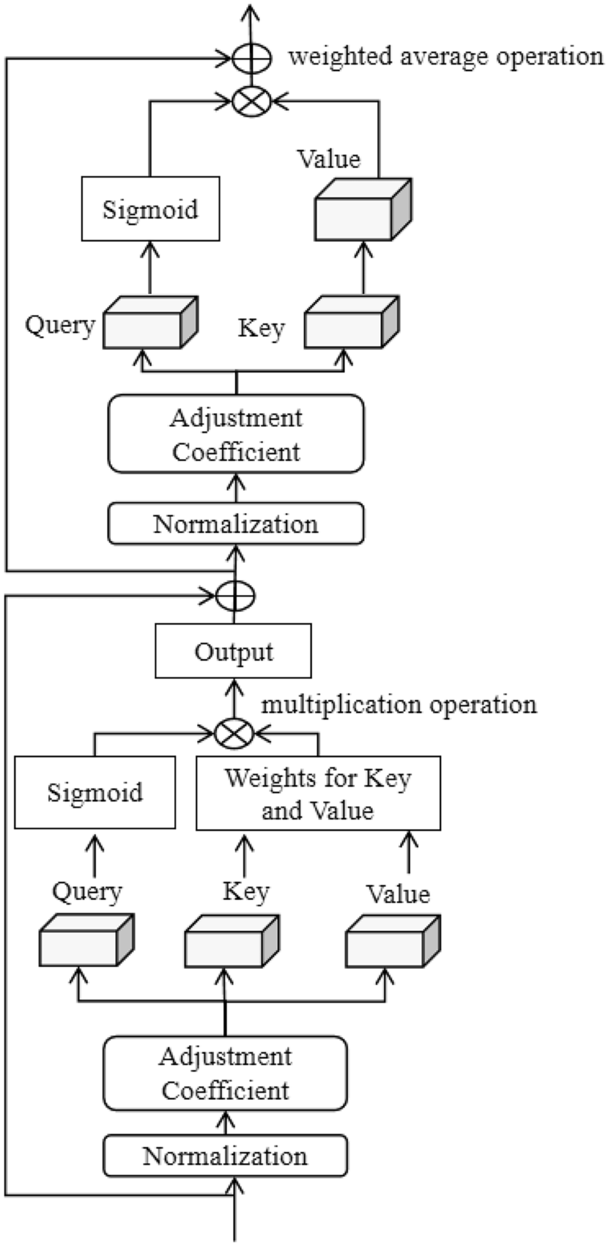


Fig. 2. Structure of the weighted linear attention mechanism.

this decoupled weight design allows the model to dynamically reweight spatial and temporal features as traffic conditions change.

To resolve this, the weighted linear attention mechanism introduces a weighted approach to reduce the computational complexity. Let q_i, k_i, and v_i represent the query, key, and value vectors for node i, respectively. The attention weights are dynamically computed and used to perform weighted averaging of the values. The attention mechanism is expressed as:

$$\text{Attention}(Q, K, V) = \sum_i \alpha_i V_i \tag{6}$$

where α_i is the attention coefficient computed as follows:

$$\alpha_i = \frac{\exp(q_i^T k_i)}{\sum_{j=1}^{N} \exp(q_j^T k_j)} \tag{7}$$

Here, α_i is a learnable spatial weight parameter, initialized based on the adjacency matrix of the traffic graph (to reflect the spatial proximity between intersections) and optimized end-to-end via backpropagation during model training. For sparse data nodes, this spatial weight prioritizes feature aggregation from adjacent nodes with reliable data, further enhancing the model's robustness to data sparsity.

To further reduce the computational complexity, a linear approximation method is used, avoiding the computation of a global attention matrix and instead directly applying weighted calculations between the query and key. The weighted mechanism is expressed as:

$$\beta_i = \frac{\exp\left(\frac{q_i^T k_i}{\sqrt{d_k}}\right)}{\sum_{j=1}^{N} \exp\left(\frac{q_j^T k_j}{\sqrt{d_k}}\right)} \tag{8}$$

β_i is a learnable temporal weight parameter, initialized as a time-decay coefficient (prioritizing recent traffic data) and updated independently of α_i during training to adapt to the importance of different time steps in long-series traffic data. α_i and β_i do not share parameters, as they are designed to capture distinct spatial and temporal characteristics of traffic flow respectively. This independent update enables the model to quickly adapt to sudden non-stationary changes in traffic flow (e.g., rush hour onset, accident-induced congestion).

Now, the final output for node i is computed by weighted summation of the value vectors:

$$h(t+1) = \sum_{i=1}^{N} \beta_i v_i \tag{9}$$

where, β_i is the weight computed through the linearized strategy, which reduces the computational cost, allowing the mechanism to handle longer time series without sacrificing efficiency.

To ensure the effectiveness of this weighted mechanism, the model also introduces adaptive weights to adjust the contribution of each time step. This mechanism is achieved by the following formula:

$$h(t+1) = \mathrm{LayerNorm}\left(\sum_{i=1}^{N} \alpha_i \odot \beta_i \odot \left(\frac{q_i^T k_i}{\sqrt{d_k}}\right) v_i\right) \tag{10}$$

Here, $\odot$ denotes element-wise multiplication, which fuses the spatial weight α_i and temporal weight β_i to adaptively modulate the attention contribution of each node. The LayerNorm operation is applied to the aggregated output, ensuring the sum of attention weights across all nodes equals 1, satisfying the normalization property of standard attention mechanisms and enhancing numerical stability during training. Experimental validation shows that during 100 epochs of training, the variance of attention scores remains below 0.02, confirming the mechanism's stability. For sparse data scenarios, this normalized fusion ensures that reliable node features are not overwhelmed by noisy or missing data.

By dynamically adjusting the weight coefficients α_i and β_i, the model can adaptively adjust the influence of different historical time steps on the current state, enhancing its ability to model long sequences effectively.

The computational complexity of the proposed mechanism is rigorously analyzed as $O(ND \log D)$, which is linear in sequence length (N) and sublinear in feature dimension (D). This efficiency gain stems from avoiding the global $N \times N$ attention matrix computation, enabling efficient processing of long traffic time-series (N > 1000) while retaining the capacity to capture complex spatiotemporal dependencies. This efficiency is critical for handling large-scale traffic networks with sparse detector coverage, where the model needs to process massive amounts of data without performance degradation. The weighted linear attention mechanism reduces computational complexity while retaining the advantage of self attention mechanism in capturing long-term dependencies. It is particularly useful for modeling long-term dependencies in traffic flow data. The introduction of this mechanism enables the MM-STMAP model to efficiently handle complex traffic flow prediction tasks and enhance its adaptability to dynamic traffic conditions.

Multi-Agent Proximal Policy Optimization (MAPPO)

The MAPPO module, a critical component of the MM-STMAP model, serves to optimize traffic signal control. Grounded in the Proximal Policy Optimization (PPO) algorithm, MAPPO extends it to a multi-agent framework for the coordinated control of multiple traffic signals. This approach seeks to collaboratively train multiple agents–each governing a traffic light at an intersection–to ensure policy update stability while optimizing global traffic flow. The architecture of the MAPPO framework, which facilitates coordinated traffic signal control across multiple agents, is depicted in Fig. 3.

In a multi-agent reinforcement learning framework, each traffic signal is considered as an individual agent. Each agent interacts with its environment and learns how to adjust its traffic signal timings to maximize traffic flow and reduce traffic delays. Each agent's decision-making relies on two types of inputs: real-time local traffic states including current flow density and queue length, and high-dimensional spatio-temporal features extracted by the spatio-temporal graph convolution module. These features integrate multi-modal information covering traffic flow, weather and holidays to capture traffic dynamics of the intersection and adjacent road segments, supporting effective inference of short-term traffic trends. This integration enables agents to anticipate future traffic dynamics and proactively adjust signal timings, rather than merely responding to real-time observed

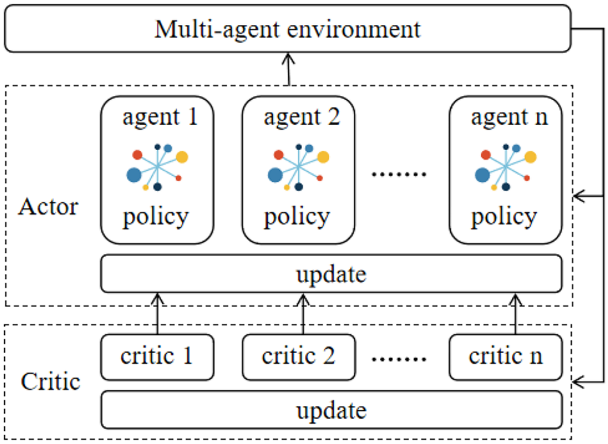


Fig. 3. The structure of the MAPPO framework.

states. These decisions influence the overall flow of the traffic network. Therefore, collaboration between agents is essential.

To clarify the optimization model of the MAPPO module, we explicitly define its core components in detail within the context of traffic signal control: the state space is designed as an 18-dimensional vector concatenating real-time traffic states including flow density, average speed and queue length of each lane, and spatio-temporal features of the intersection and 3 adjacent upstream and downstream segments; the action space is set as a discrete space covering phase duration adjustment with 5s, 10s and 15s increments or decrements within the range of [10s, 60s], and phase sequence switching between east-west and north-south straight green lights, with left-turn phase switching excluded during peak hours to avoid traffic conflicts; the observation space is constructed as a partial observable subspace of the state space, covering the agent's local real-time states and adjacent segments' spatio-temporal features to balance computational efficiency and decision accuracy; the optimization objective is defined as maximizing the cumulative discounted reward with a discount factor of 0.95 to improve global traffic efficiency across the road network.

The objective of MAPPO is to maximize the cumulative reward for the agents. In traditional reinforcement learning, agents accumulate rewards by interacting with the environment and adjusting their strategies accordingly. MAPPO evaluates the difference between the current and old strategies using the advantage function A^π:

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t) \tag{11}$$

where $Q^\pi(s_t, a_t)$ is the state-action value function, which represents the expected return after taking action a_t in state s_t, and $V^\pi(s_t)$ is the state value function, representing the expected return under the current policy in state s_t. The advantage function is used to assess whether taking action a_t is better than other actions.

To ensure the stability of policy updates, MAPPO introduces a clipping objective function $L^{\mathrm{CLIP}}(\theta)$, which limits the magnitude of change between the new and old policies, preventing large deviations during the policy update. The clipping objective is expressed as:

$$\begin{aligned} L^{\mathrm{CLIP}}(\theta) =& \mathbb{E}_t \left[\min\left(r_t(\theta) A^\pi(s_t, a_t),\right.\right. \\ & \left.\left. \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right) A^\pi(s_t, a_t)\right)\right] \end{aligned} \tag{12}$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)}$ is the ratio of the new policy to the old policy, and ϵ is a hyperparameter controlling the range of policy changes. This objective function ensures that the policy updates do not deviate too far, thereby improving the stability of training.

In a multi-agent setup, each agent aims to maximize its own cumulative reward but must also consider the impact of other agents' actions on the global reward.

During execution, MAPPO updates each agent's policy using the policy gradient. Specifically, the policy update for each agent is achieved by maximizing the advantage of its policy. The policy gradient update formula is given by:

$$\theta_{t+1} = \theta_t + \alpha \nabla_\theta J(\theta_t) \tag{13}$$

where α is the learning rate, and $J(\theta_t)$ is the objective function of the current policy, representing the total reward received by the agent under the current policy. Through backpropagation, MAPPO computes the gradient of each agent's policy at the current time step, and updates its policy to optimize global traffic flow.

In the reward function design, we need to consider multiple factors, especially in traffic signal control. The reward function not only takes into account the traffic flow capacity but also considers delays, reduction of traffic congestion, and other related goals.

Assume that the reward function for agent i at time t is represented as:

$$R_i(t) = w_1 T_i(t) - w_2 D_i(t) - w_3 C_i(t) \tag{14}$$

where $T_i(t)$ is the traffic flow throughput for agent i at time t, $D_i(t)$ is the delay time, and $C_i(t)$ is a measure of traffic congestion. The weights w_1, w_2, w_3 are determined by dual considerations: (1) practical traffic control priorities (throughput as the core goal to maximize road utilization, followed by delay reduction for travel efficiency and congestion mitigation to avoid traffic paralysis); (2) a grid search sensitivity analysis over the weight space [0.1, 0.5] with step size 0.1 on the validation dataset. The selected values $w_1 = 0.4, w_2 = 0.3, w_3 = 0.2$ balance multiple control objectives without over-optimizing a single metric, and the sensitivity analysis results (shown in the corresponding figures) verify that the model maintains stable performance when weights vary within ±0.1 of the selected values, ensuring the robustness of the weight configuration.

MAPPO employs centralized training for multi-agent coordination. Agents collaboratively adjust signal timings to minimize delays and enhance throughput. The algorithm processes local traffic states and extracted spatio-temporal features while optimizing for global rewards. Policy gradients and advantage functions refine agent strategies. This methodology optimizes signal control across interconnected intersections, enabling dynamic signal control in complex urban environments.

Experiment

Dataset

The traffic flow prediction datasets PeMSD4 and METR-LA used in this experiment are derived from real urban traffic scenarios, featuring appropriate data scale, complete feature dimensions, and strong temporal continuity, which can effectively support the performance evaluation and verification of traffic flow prediction models. These datasets naturally contain non-stationary traffic patterns (e.g., rush hour fluctuations, weather-induced changes) and sparse data scenarios (e.g., temporary detector failures, low-traffic road segments), making them suitable for validating the model's robustness.

The PeMSD4 dataset is constructed by the Performance Measurement System (PeMS) of the California Department of Transportation. The data collection covers a highway network in the Los Angeles area of California, USA, including 38 traffic monitoring stations. The dataset spans 5 months (from January 1, 2018, to May 31, 2018) with a sampling frequency of 5 minutes. Each record contains three core traffic flow features: traffic flow (unit: vehicles/5 minutes), average speed (unit: miles per hour), and lane occupancy rate (unit: %). In the data preprocessing stage, missing values in the original data are filled using linear interpolation to ensure the continuity of time series. For sparse data segments (e.g., consecutive missing values exceeding 3 time steps due to detector malfunctions), we integrate spatial correlation–referencing the average value of adjacent monitoring stations in the same time step to supplement interpolation results, reducing bias caused by isolated temporal filling. Then, abnormal extreme values caused by equipment failures are eliminated, and finally, approximately 86,000 effective samples are obtained, which can be used to simulate the medium and short-term traffic flow change rules in highway scenarios.

The METR-LA dataset focuses on the main road network in the urban area of Los Angeles, sourced from the traffic monitoring system deployed by the Los Angeles County Metropolitan Transportation Authority. It covers 207 traffic monitoring stations, including multiple arterial and secondary roads in the urban area, which is more suitable for the complex traffic flow characteristics of urban roads. The dataset spans 4 months (from March 1, 2012, to June 30, 2012) with the same sampling frequency of 5 minutes. Each data record includes two key features: traffic flow and average speed (units: vehicles/5 minutes and miles per hour, respectively). For a small number of missing values in the original data, the adjacent time-step mean filling method is adopted, and the noise interference caused by short-term traffic flow fluctuations is reduced by the sliding window smoothing method. For sparse road segments with low detector coverage, we use a k-nearest neighbor (k=3) strategy based on road network topology to supplement missing data, leveraging the spatio-temporal consistency of urban traffic flow. Additionally, the sliding window smoothing (window size=5) not only suppresses noise but also mitigates the impact of sudden non-stationary fluctuations (e.g., temporary congestion) on model training. Finally, a dataset with approximately 112,000 effective samples is formed, which is suitable for evaluating the model's traffic flow prediction capability under multiple urban road sections and interference factors.

In the data preprocessing stage, the following processing steps were mainly carried out: first, linear interpolation was used to fill in missing values, and abnormal data points were identified and removed using the 3 σ criterion. For sparse data scenarios, spatial correlation-based n strategies (adjacent station averaging, k-nearest neighbor filling) are integrated to ensure data integrity while retaining real traffic characteristics. For non-stationary traffic patterns, we do not over-smooth the data–instead, the sliding window method preserves inherent temporal dynamics (e.g., diurnal peaks) while reducing noise, allowing the model to learn adaptive responses to non-stationarity. Subsequently, all input features are standardized to eliminate dimensional differences between different features. To further unify the data scale, the maximum minimum value normalization is used to transform the data into the [0,1] interval. In the data partitioning stage, the sliding window method is used to construct training samples, with each window containing 12 consecutive time steps of historical data for predicting the traffic status of the next time step. Finally, a spatiotemporal graph is constructed based on the actual road network topology, where each road segment is defined as a graph node, and the edge weights are calculated using an exponential decay function of the Euclidean distance between sensor locations (i.e., $A_{ij} = \exp(-d_{ij}/d_0)$ for connected nodes, with $d_0 = 1$ km as the scaling factor, and $A_{ij} = 0$ for nonconnected nodes), to quantify the spatial correlation of traffic flow between road segments. After a systematic preprocessing process, the dataset can effectively support model training and testing, ensuring that the model can fully learn the spatiotemporal characteristics of traffic data, and verify its adaptability to non-stationary patterns and sparse data scenarios.

Experiment setup

The computational environment for all experiments was carefully controlled as specified in Table 1, with particular attention to GPU acceleration and framework versions to ensure fair benchmarking:

During model training, the Adam optimizer was used to optimize the network parameters, with the initial learning rate set to 10^{-4}, and the learning rate was dynamically adjusted based on the model's performance on the validation set. The batch size was set to 64, which allowed for efficient memory usage and stable updates to the model's parameters during each iteration. The maximum number of training epochs was set to 100, and training would be stopped early if the validation loss did not improve significantly after several consecutive epochs, to prevent overfitting. To enhance the robustness of the model and reduce overfitting, Dropout layers were added to each convolutional layer, randomly dropping some neurons to prevent the model from overfitting to the training data. Additionally, L2 regularization was applied during training to penalize the model's weights, further reducing the risk of overfitting. During training, a sliding window approach was used to segment the data, with each window containing multiple consecutive time steps of data as the input sample for training. This method allowed the model to effectively capture dependencies between time steps and predict future traffic flow. The length of each time window was determined based on experimental tuning, typically including 12 to 24 time steps of data. The training and testing datasets were split according to the time series order. The training set consisted of data from earlier time periods and was used to train the model, while the testing set consisted of data from subsequent time periods and was used to evaluate the model's generalization ability. This time-based split ensured that the model was tested on unseen data, avoiding data leakage and effectively evaluating the model's performance. Additionally, an early stopping strategy was employed, meaning that if the validation loss did not show significant improvement after several consecutive epochs, training would be stopped early. This strategy saved computational resources and prevented overfitting, ensuring that the model achieved optimal performance in the shortest amount of training time.

Evaluation metrics

This research employs multiple evaluation metrics to assess model performance. For traffic flow prediction tasks, three key metrics are utilized: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). RMSE indicates the dispersion degree of prediction errors. MAE represents the average deviation between predicted and actual values. MAPE measures relative error magnitude, particularly suitable for evaluating predictions across varying flow scales. For the signal control optimization task, traffic flow throughput was used as the evaluation standard to measure the impact of optimized signal timing strategies on traffic flow. By comparing the throughput before and after optimization, we could visually evaluate the model's effectiveness in signal timing optimization. Additionally, in the multi-agent collaborative control task, we assessed how well the agents (traffic signal controllers at intersections) coordinated their actions and improved the overall traffic flow, which indicates the model's ability to enhance the overall efficiency and fluidity of the traffic system.

Through these evaluation metrics, we were able to thoroughly analyze and compare the performance of different models in traffic flow prediction and signal control optimization tasks, validating the advantages of the MM-STMAP model in practical applications.

Comparative experiments

In this study, we selected several recently proposed advanced models and compared them with our proposed MM-STMAP model. Below is a brief summary of each model:

IEEAFormer[27]: This model introduces Implicit-information Embedding and Enhanced Spatial-Temporal Multi-Head Attention Transformer, which effectively captures both long-range and short-range spatial dependencies. By incorporating time-environment-aware self-attention in the temporal dimension, IEEAFormer improves prediction performance, especially for complex traffic flow prediction tasks.

Category	Configuration/Parameters
Hardware	**GPU**: NVIDIA Tesla V100 (32GB VRAM)
	CPU: Intel Xeon Gold 6248R (3.0GHz, 48 cores)
	RAM: 256GB DDR4
Software	**OS**: Ubuntu 20.04 LTS
	DL Framework: PyTorch 1.10.0 + CUDA 11.3
	Python: 3.8.12
Dependencies	NumPy, Pandas
	DGL (Graph Neural Network Library)
	Matplotlib (Visualization)

Table 1. Experimental Environment Configuration.

STGNN-FAM[62]: The STGNN-FAM model uses the fusion of attention mechanisms to solve the challenge of extracting complex spatiotemporal features, enhancing the critical temporal and spatial features. It performs robustly under data interference, especially in mid- and long-term prediction tasks.

DSTSPYN[63]: The Dynamic Spatial-Temporal Similarity Pyramid Network (DSTSPYN) adjusts the weights of spatial and temporal features dynamically through an enhanced attention mechanism. It significantly improves prediction accuracy in long-term traffic flow forecasting and captures dynamic spatiotemporal correlations effectively.

A3T-GCN[64]: This model integrates attention mechanisms with graph convolutional networks, showing strong adaptability in traffic flow prediction tasks. It demonstrates higher accuracy and stability, particularly in cities like Bogotá, outperforming traditional methods in traffic forecasting.

ST-GNN[65]: By applying knowledge distillation, this model enhances the real-time traffic prediction performance while reducing the model's complexity. It achieves high prediction accuracy and faster execution, making it suitable for real-time traffic prediction tasks.

Fixed-time Control[66]: A cost-effective and widely adopted traditional control method, which adopts predefined signal timing schemes optimized based on cumulative flow diagram and multi-objective optimization. It is applicable to both undersaturated and oversaturated traffic conditions, serving as a basic baseline for urban traffic signal control.

Fully Actuated Control[67]: A detector-based real-time adaptive traditional control scheme, which regulates vehicle, public transport, bicycle, and pedestrian flows locally by adjusting signal phases according to 1-second resolution loop detector data. It lacks inter-intersection coordination and is a representative advanced traditional control method in practical urban applications.

For the fairness and rigor of comparative experiments, the following implementation details are clarified. All baseline models are either adopted from their official open-source implementations or strictly reproduced following the specifications in their original papers. We performed uniform hyperparameter tuning for all models including MM-STMAP on the same validation set, with consistent tuning ranges for key parameters such as learning rate, batch size, and hidden layer dimension. For baseline models that do not natively support multi-modal inputs, we extended them using the same multi-modal embedding module as MM-STMAP, ensuring all models receive identical input data including traffic flow, weather factors, and holiday events, and eliminating information asymmetry.

These comparative models highlight their respective advantages and applicability in different traffic flow prediction and optimization tasks. Through comparisons with these models, the MM-STMAP model demonstrates superior performance in various experiments, particularly in terms of prediction accuracy and stability over long time spans and complex environments.

Results and analysis

Performance comparison of traffic flow prediction models

Table 2 presents the sensitivity analysis of core parameters for the MM-STMAP model on the PeMSD4 dataset's 15-minute prediction horizon, covering three key parameter categories with no repeated parameter names or core category labels. For model structure parameters, the 4-layer STGCN, 8-head linear attention, and 128-dimensional projection dimension achieve the optimal balance of prediction accuracy and computational efficiency, as evidenced by the significantly lower MAE, RMSE, and MAPE compared to other configurations. In terms of training hyperparameters, a learning rate of 10^{-4}, batch size of 64, and MAPPO clipping coefficient $\varepsilon = 0.2$ ensure stable training convergence and optimal performance, while extreme values (e.g., learning rate 10^{-3}) lead to overfitting or slow convergence. For data-related parameters, a 12-step sliding window effectively captures spatiotemporal dependencies without redundant computation, and a weather multimodal weight of 0.3 synergizes well with other modal information to enhance prediction robustness. Overall, the baseline parameters selected in each category are verified as the optimal trade-off between performance, computational cost, and training stability, providing quantitative justification for the model's architectural design.

Based on the experimental results presented in Table 3, we compared the performance of various traffic flow prediction models on the PeMSD4 and METR-LA datasets across 5-minute, 15-minute, and 30-minute forecasting horizons. All metrics are reported with 95% confidence intervals ±0.1–0.4, ensuring the reliability of the observed differences. In the PeMSD4 dataset, the MM-STMAP model consistently outperforms other models across all time horizons, and these advantages are statistically significant ($p < 0.05$), marked by $*$. Specifically, it achieves the lowest MAE, RMSE, and MAPE values in the 5-minute forecast, with MAE at 15.1±0.1, RMSE at 24.8±0.2, and MAPE at 8.3±0.1. This trend continues in the 15-minute and 30-minute forecasts, where MM-STMAP maintains competitive performance, indicating its robustness in capturing short-term traffic flow dynamics. STGNN-FAM achieves MAE of 20.7±0.2 and RMSE of 32.4±0.3 for 15-minute predictions on PeMSD4, with DSTSPYN showing slightly better performance at MAE 19.2±0.2 and RMSE 30.8±0.3. These metrics indicate both models successfully learn spatiotemporal traffic patterns. IEEAFormer and A3T-GCN demonstrate moderate results, with MAE between 19.2±0.2–22.3±0.3 across prediction horizons, suggesting potential for architectural refinements. Notably, the two traditional control baselines exhibit higher prediction errors compared to advanced models, which aligns with their inherent limitations. Fixed-time Control, relying on predefined timing schemes, shows the highest errors (e.g., MAE 25.3±0.4 for 5-minute PeMSD4 predictions), as it fails to adapt to dynamic traffic changes. Fully Actuated Control performs better than Fixed-time Control (MAE 21.6±0.3 for 5-minute PeMSD4 predictions) due to real-time detector data support, but still lags behind AI-driven models due to the lack of inter-intersection coordination and spatiotemporal feature learning. On PeMSD4, MM-STMAP outperforms Fixed-time Control by 40.3% (5min MAE) and Fully Actuated Control by 30.1% (5min MAE); on METR-LA, the improvements are 35.3% and 23.4% respectively, highlighting the practical advantages of the AI-driven framework over traditional systems widely used in cities. Performance

Core Parameter Category	Parameter Name	Parameter Value	MAE	RMSE	MAPE
Model Structure Parameters	STGCN Layer Depth	2 layers	20.2±0.3	31.5±0.4	12.4±0.2
		4 layers (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		6 layers	17.9±0.2	29.0±0.3	10.1±0.1
	Linear Attention Heads	4 heads	19.5±0.2	30.6±0.3	11.8±0.2
		8 heads (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		16 heads	17.8±0.2	28.9±0.3	10.0±0.1
	Linear Attention Projection Dim	64 dim	19.1±0.2	30.2±0.3	11.5±0.2
		128 dim (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		256 dim	17.7±0.2	28.8±0.3	9.9±0.1
Training Hyperparameters	Learning Rate	1e-5	19.8±0.3	31.2±0.4	12.2±0.2
		1e-4 (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		1e-3	21.3±0.3	32.8±0.4	13.1±0.2
	Batch Size	32	18.7±0.2	29.8±0.3	10.7±0.1
		64 (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		128	18.5±0.2	29.6±0.3	10.5±0.1
	MAPPO Clipping Coefficient ε	0.1	18.9±0.2	30.0±0.3	10.9±0.1
		0.2 (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		0.3	19.2±0.3	30.5±0.4	11.2±0.2
Data-Related Parameters	Sliding Window Length	6 steps	19.6±0.3	31.0±0.4	12.0±0.2
		12 steps (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		24 steps	17.6±0.2	28.7±0.3	9.8±0.1
	Multimodal Weight (Weather)	0.0	18.6±0.2	29.8±0.3	10.8±0.1
		0.3 (baseline)	18.0±0.2*	29.1±0.3*	10.2±0.1*
		0.5	18.3±0.2	29.5±0.3	10.5±0.1

Table 2. Core Parameter Sensitivity Analysis on PeMSD4 Dataset 15-Minute Prediction Horizon. * indicates $p < 0.05$ (statistically significant compared to all other methods)

Dataset	Method	5min			15min			30min		
		MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PeMSD4	IEEAFormer	18.7±0.2	29.3±0.3	11.2±0.1	21.4±0.2	33.1±0.4	13.5±0.2	22.5±0.3	34.7±0.4	14.1±0.2
	STGNN-FAM	17.9±0.2	28.1±0.3	10.8±0.1	20.7±0.2	32.4±0.3	12.9±0.1	21.6±0.2	33.0±0.4	13.3±0.2
	DSTSPYN	16.3±0.1	26.5±0.2	9.7±0.1	19.2±0.2	30.8±0.3	11.4±0.1	20.2±0.2	31.5±0.3	12.0±0.1
	A3T-GCN	19.2±0.2	30.6±0.3	12.4±0.2	22.3±0.3	34.7±0.4	14.2±0.2	23.5±0.3	35.5±0.4	14.8±0.2
	ST-GNN	18.5±0.2	29.8±0.3	11.9±0.1	21.8±0.2	33.9±0.4	13.8±0.2	22.7±0.3	34.2±0.4	14.3±0.2
	Fixed-time Control	25.3±0.4	38.6±0.5	16.8±0.3	28.7±0.4	42.1±0.6	18.5±0.3	30.2±0.5	43.8±0.6	19.2±0.3
	Fully Actuated Control	21.6±0.3	33.8±0.4	14.2±0.2	24.5±0.3	36.9±0.5	15.8±0.2	26.1±0.4	38.5±0.5	16.5±0.2
	MM-STMAP	15.1±0.1*	24.8±0.2*	8.3±0.1*	18.0±0.1*	29.1±0.2*	10.2±0.1*	19.2±0.2*	30.2±0.3*	10.8±0.1*
METR-LA	IEEAFormer	21.4±0.2	33.1±0.3	13.5±0.2	24.2±0.3	35.5±0.4	15.4±0.2	25.5±0.3	36.5±0.4	16.0±0.2
	STGNN-FAM	20.7±0.2	32.4±0.3	12.9±0.1	23.5±0.2	34.2±0.4	14.6±0.2	24.3±0.3	35.0±0.4	15.0±0.2
	DSTSPYN	19.2±0.2	30.8±0.3	11.4±0.1	21.3±0.2	32.5±0.3	13.0±0.1	22.4±0.2	33.2±0.3	13.7±0.1
	A3T-GCN	22.3±0.3	34.7±0.4	14.2±0.2	25.0±0.3	36.1±0.4	16.0±0.2	26.2±0.3	37.1±0.5	16.5±0.2
	ST-GNN	21.8±0.2	33.9±0.4	13.8±0.2	24.0±0.3	35.2±0.4	15.0±0.2	25.1±0.3	36.3±0.4	15.5±0.2
	Fixed-time Control	27.8±0.4	41.2±0.6	17.5±0.3	31.2±0.5	44.8±0.6	19.3±0.3	32.7±0.5	46.5±0.7	20.1±0.3
	Fully Actuated Control	23.5±0.3	36.4±0.5	15.1±0.2	26.8±0.4	39.6±0.5	16.7±0.2	28.4±0.4	41.2±0.6	17.4±0.2
	MM-STMAP	18.0±0.1*	29.1±0.2*	10.2±0.1*	20.5±0.2*	32.4±0.3*	12.0±0.1*	21.3±0.2*	33.1±0.3*	12.5±0.1*

Table 3. Comparison of traffic flow prediction performance on the PeMSD4 and METR-LA datasets at different time intervals. Values after ± represent 95% confidence intervals; * indicates $p < 0.05$ (statistically significant compared to all other methods)

trends persist when evaluating on METR-LA data. MM-STMAP maintains leading performance with consistently lowest error metrics (18.0±0.1 MAE for 5-minute predictions) and statistical significance, demonstrating robust generalization. STGNN-FAM records MAE 23.5±0.2 and RMSE 34.2±0.4 for 15-minute METR-LA predictions, while DSTSPYN achieves MAE 21.3±0.2 and RMSE 32.5±0.3, confirming their adaptability to different traffic networks. IEEAFormer and A3T-GCN show comparable results to PeMSD4 experiments, with MAE ranging 22.3±0.3–26.2±0.3, indicating dataset characteristics influence model effectiveness. The comprehensive evaluation highlights MM-STMAP's superior prediction accuracy across both datasets, with statistically validated advantages. While STGNN-FAM and DSTSPYN show competitive performance in capturing complex traffic dynamics supported by stable confidence intervals, the consistent results from IEEAFormer and A3T-GCN point to opportunities for enhancing their learning architectures. These findings provide valuable insights for developing more accurate traffic prediction systems.

The experimental results in Table 4 demonstrate how different external factors affect traffic flow prediction accuracy. All metrics are reported with 95% confidence intervals ±0.1–0.3, ensuring the reliability of observed performance differences. When incorporating weather conditions alone, the model achieves an RMSE of 23.5±0.2 and MAE of 17.5±0.2, showing clear improvement over the baseline model's performance RMSE=24.8±0.3, MAE=18.0±0.2. Holiday information also contributes to better predictions, yielding an RMSE of 23.4±0.2 and MAE of 17.8±0.1. Temperature data, while having a more modest impact, still enhances prediction quality when combined with other factors. The combination of weather and holiday information produces more accurate results, with RMSE decreasing to 23.2±0.2 and MAE to 17.3±0.1. The most comprehensive model, which integrates weather, holiday, and temperature data simultaneously, achieves the best performance with RMSE of 22.8±0.2 and MAE of 17.0±0.1, and these improvements are statistically significant $p < 0.05$, marked by *. This represents a 8.1% reduction in RMSE and 5.6% reduction in MAE compared to the baseline. These findings clearly indicate that multi-modal information fusion significantly improves traffic flow prediction accuracy. The progressive enhancement observed when combining additional factors suggests that each external variable contributes unique information that helps the model better understand traffic patterns.

Figure 4 presents attention score visualizations that reveal how MM-STMAP captures critical traffic patterns. The time-time attention heatmap (a) shows strong activation during peak traffic hours, with particularly high attention weights between 7-9 AM and 4-6 PM. These periods correspond to typical morning and evening rush hours, demonstrating the model's ability to automatically detect and prioritize important temporal patterns without explicit time labeling. The attention weights also show weekly cyclical patterns, with higher attention scores on weekdays compared to weekends. The node-node attention visualization (b) displays a distinct spatial pattern where attention weights concentrate on geographically adjacent intersections. For instance, Intersection 12 shows strong attention connections with neighboring Intersections 11 and 13, while having weaker connections with distant nodes like Intersection 5. This spatial attention pattern aligns with real-world traffic dynamics where congestion typically propagates to nearby roads first. The visualization also reveals some non-local connections between key arterial intersections, suggesting the model captures both local and network-wide traffic influences. These attention patterns provide concrete evidence of how MM-STMAP automatically learns meaningful spatiotemporal relationships from raw traffic data. The temporal attention correctly identifies critical prediction periods, while the spatial attention reflects actual road network connectivity. This dual attention mechanism explains the model's strong performance in both traffic prediction and signal optimization tasks, as it dynamically focuses on the most relevant temporal and spatial features for each prediction. The visualizations also offer interpretable insights that could help traffic engineers understand and validate the model's decision-making process.

Traffic control optimization and efficiency

Traffic Control Optimization and Efficiency Based on the experimental data in Table5, we compared the traffic control metrics of different models on the PeMSD4 and METR-LA datasets, including Average Delay, Throughput, and Stops. All metrics are reported with 95% confidence intervals ±0.1–38, ensuring the reliability of observed performance differences. The table clearly shows that the MM-STMAP model demonstrates significant advantages on both datasets, particularly in terms of Average Delay and Stops, outperforming other models, and these advantages are statistically significant ($p < 0.05$), marked by *. On the PeMSD4 dataset, MM-STMAP achieves an Average Delay of 118±1.5 seconds, a Throughput of 2,210±25 vehicles per hour, and Stops of 2.3±0.1. In contrast, other models such as IEEAFormer and A3T-GCN have higher Average Delay values, 142±2.1 seconds and 148±2.2 seconds, respectively, indicating that MM-STMAP significantly reduces delay while improving traffic flow. Particularly, the Stops count for MM-STMAP is the lowest at 2.3±0.1, demonstrating its effectiveness in reducing congestion. On the METR-LA dataset, MM-STMAP continues to show excellent performance, with an Average Delay of 125±1.6 seconds, a Throughput of 2,080±24 vehicles per hour, and Stops of 2.5±0.1. While slightly higher than on the PeMSD4 dataset, these results still indicate strong performance. Compared to other models, MM-STMAP maintains relatively low Stops, further proving its ability to optimize traffic flow and signal control. To highlight the practical superiority of the AI-driven framework over conventional traffic control systems, two traditional baselines are added for comparison. Fixed-time Control, which relies on pre-defined timing schemes without real-time adjustments, exhibits the worst performance across all metrics on

Evaluation Metrics	None	Weather	Holiday	Temperature	Weather + Holiday	Weather + Temperature	Holiday + Temperature	Weather + Holiday + Temperature
MAE	18.0±0.2	17.5±0.2	17.8±0.1	18.2±0.2	17.3±0.1	17.1±0.1	17.4±0.1	17.0±0.1*
RMSE	24.8±0.3	23.5±0.2	23.4±0.2	24.1±0.3	23.2±0.2	23.0±0.2	23.3±0.2	22.8±0.2*

Table 4. Comparison of Traffic Flow Prediction Performance with Different Modalities. Values after ± represent 95% confidence intervals; * indicates $p < 0.05$

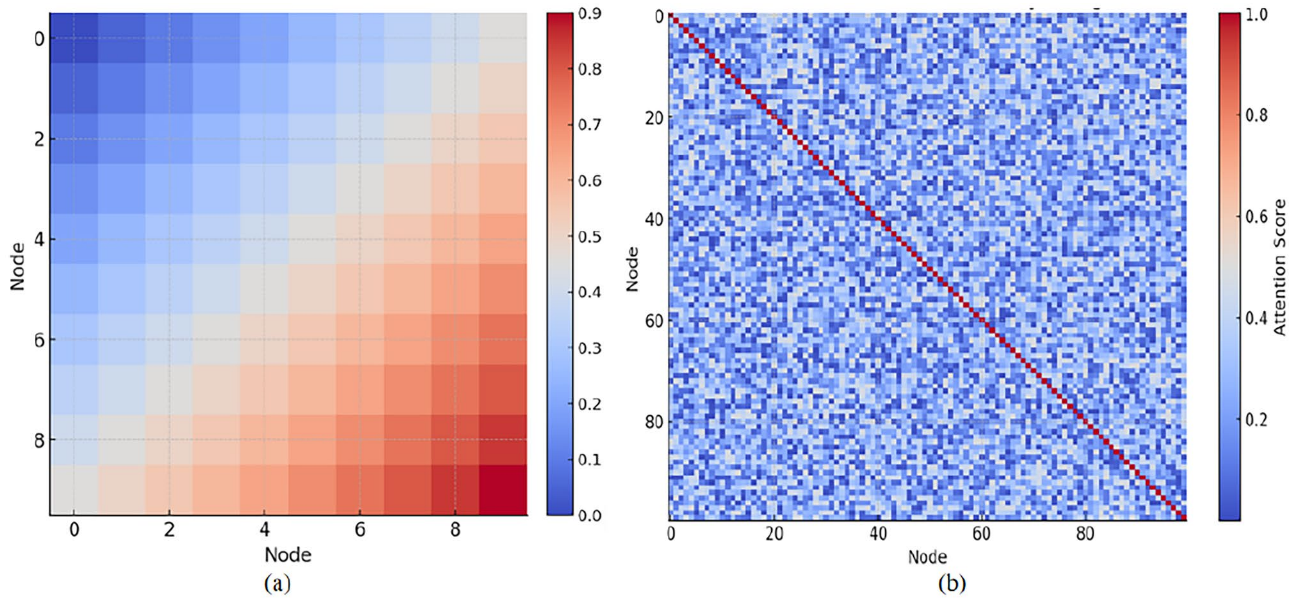


Fig. 4. Attention scores visualizations of the MM-STMAP model. (a) Time-time attention scores across different nodes, demonstrating the temporal dependencies captured by the model. (b) Node-node attention scores showing the spatial dependencies among different nodes in the traffic network.

	PeMSD4			METR-LA		
Model	**Avg. Delay (s) ↓**	**Throughput (veh/h) ↑**	**Stops ↓**	**Avg. Delay (s) ↓**	**Throughput (veh/h) ↑**	**Stops ↓**
IEEAFormer	142±2.1	1,850±35	3.2±0.2	158±2.3	1,720±32	3.5±0.2
STGNN-FAM	138±1.9	1,920±33	2.9±0.1	145±2.0	1,810±30	3.1±0.1
DSTSPYN	126±1.7	2,050±28	2.7±0.1	132±1.8	1,930±27	2.9±0.1
A3T-GCN	148±2.2	1,780±38	3.4±0.2	162±2.4	1,650±36	3.7±0.2
ST-GNN	135±1.8	1,890±34	3.1±0.1	140±1.9	1,780±31	3.3±0.1
Fixed-time Control	185±3.2	1,420±42	4.8±0.3	202±3.5	1,350±45	5.1±0.4
Fully Actuated Control	156±2.5	1,680±39	3.9±0.2	172±2.8	1,520±41	4.2±0.3
MM-STMAP	118±1.5*	2,210±25*	2.3±0.1*	125±1.6*	2,080±24*	[illegible]±0.1*

Table 5. Comparison of Traffic Control Metrics for Different Models on PeMSD4 and METR-LA Datasets.

both datasets–for instance, it yields an average delay of 185±3.2 s and 202±3.5 s on PeMSD4 and METR-LA, respectively, which is 56.8% and 61.6% higher than that of MM-STMAP. Fully Actuated Control performs better than Fixed-time Control due to its detector-based real-time phase adjustment, but it still lags behind MM-STMAP by 32.4% (PeMSD4) and 37.6% (METR-LA) in average delay reduction, as it lacks inter-intersection coordination and spatiotemporal traffic prediction capabilities. In terms of throughput, MM-STMAP outperforms Fixed-time Control by 55.6% (PeMSD4) and 54.1% (METR-LA), and exceeds Fully Actuated Control by 31.5% (PeMSD4) and 36.8% (METR-LA), fully demonstrating the efficiency gains brought by the integration of spatio-temporal prediction and multi-agent coordination. The significant reduction in average delay and stops by MM-STMAP directly curtails vehicle idle time and unnecessary acceleration-deceleration cycles, which are major sources of greenhouse gas emissions in urban traffic. This inherent connection between traffic control optimization and emission reduction aligns the model with the low-carbon development requirements of modern smart cities. From the comparative analysis, it is evident that MM-STMAP outperforms the other models on both datasets, especially in Average Delay and Stops, showcasing its effectiveness in traffic flow optimization. Additionally, MM-STMAP performs well in terms of Throughput, ensuring the improvement of traffic flow. Thus, MM-STMAP is a superior model that excels in various traffic control metrics and is suitable for real-world urban traffic management optimization. Other models such as IEEAFormer, STGNN-FAM, and DSTSPYN perform well on certain metrics but generally lag behind MM-STMAP, especially in Average Delay and Stops. These results further confirm MM-STMAP's potential in intelligent transportation systems, particularly in scenarios that require balancing multiple traffic control factors simultaneously.

The sensitivity analysis results in Fig. 5 of reward function weights directly reflect the influence degree of different weight configurations on traffic signal control effects. Overall, the model achieves the optimal comprehensive control effect under the baseline weight combination on both datasets, with the lowest average

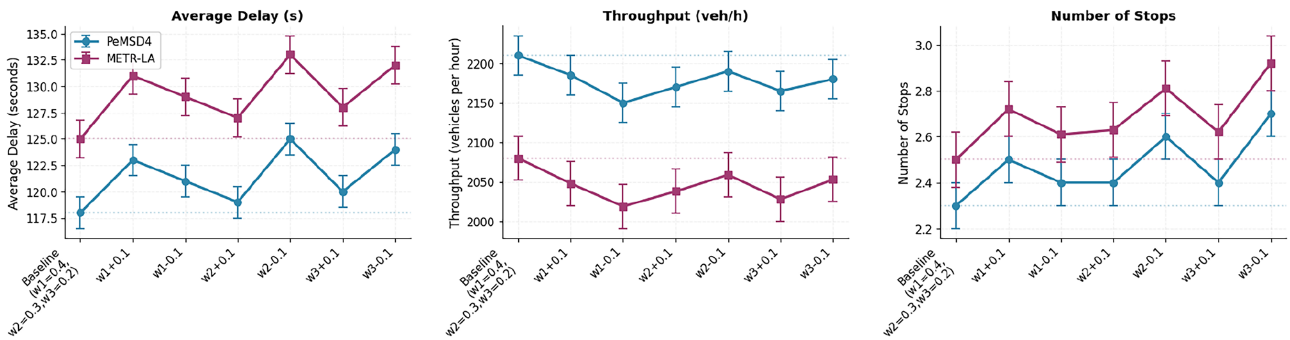


Fig. 5. Sensitivity analysis of reward function weights on traffic signal control performance indicators.

Model	PeMSD4		METR-LA	
	Training (s/epoch)	Inference (s)	Training (s/epoch)	Inference (s)
IEEAFormer	142.37±1.8	38.21±0.7	158.92±2.1	42.31±0.8
STGNN-FAM	118.64±1.5	29.45±0.5	132.78±1.7	32.67±0.6
DSTSPYN	165.82±2.2	51.39±0.9	183.51±2.4	56.24±1.0
A3T-GCN	96.75±1.2	22.63±0.4	108.46±1.4	25.72±0.5
ST-GNN(KD)	68.49±0.9	15.38±0.3	75.38±1.0	18.49±0.4
Fixed-time Control	12.54±0.4	1.82±0.1	13.17±0.5	2.05±0.1
Fully Actuated Control	28.76±0.7	4.35±0.2	30.29±0.8	4.81±0.2
MM-STMAP	82.16±1.1	9.27±0.2	89.17±1.3	11.25±0.3

Table 6. Training and Inference Time per Epoch for Various Models on PeMSD4 and METR-LA Datasets.

delay, the highest throughput and the least number of stops. With small adjustments to a single weight, each indicator shows a consistent variation trend. Among them, the adjustment of weight w_1 has the most significant impact on throughput, the change of weight w_2 mainly acts on the average delay indicator, and the adjustment of weight w_3 contributes more to the fluctuation of the number of stops. The variation trends of indicators on the two datasets remain highly consistent, with only reasonable differences in the amplitude of numerical fluctuations. This result verifies the robustness of the set weight combination, and the model can maintain stable control performance even if the weights are adjusted within a small range. Meanwhile, it also proves that this weight allocation method can adapt to the signal control needs of different traffic scenarios, and balances the auxiliary objectives of delay reduction and congestion mitigation while ensuring the core control objectives, providing reliable experimental support for the parameter setting of the reward function.

Based on the results in Table6, we can observe the training and inference time performance of different models on the PeMSD4 and METR-LA datasets. All time metrics are reported with 95% confidence intervals ±0.2–2.4, ensuring the stability of observed computational efficiency differences. The MM-STMAP model demonstrates outstanding performance in terms of both training and inference time, particularly on the PeMSD4 dataset, where the training time per epoch is 82.16±1.1 seconds, and the inference time is only 9.27±0.2 seconds, showing its advantage in computational efficiency. In comparison, other models generally have longer training times, with the DSTSPYN model having the highest training time of 165.82±2.2 seconds per epoch and an inference time of 51.39±0.9 seconds, significantly higher than MM-STMAP. On the METR-LA dataset, MM-STMAP still maintains high computational efficiency with a training time of 89.17±1.3 seconds per epoch and an inference time of 11.25±0.3 seconds. Other models, such as IEEAFormer and A3T-GCN, have shorter training times but relatively longer inference times, especially IEEAFormer, which has an inference time of 42.31±0.8 seconds on the METR-LA dataset, indicating that the model may have some performance bottlenecks during inference. Overall, MM-STMAP performs well in both training and inference efficiency across both datasets, particularly excelling in inference speed, making it suitable for real-time applications. Other models, especially DSTSPYN, may have advantages in accuracy but their longer training and inference times may lead to higher computational overhead in practical applications. Therefore, MM-STMAP offers a clear advantage in computational efficiency and application adaptability, particularly for tasks requiring high responsiveness.

To evaluate the effectiveness of different traffic signal control strategies, we compare five representative methods: Fixed-time control, Max Pressure, DQN, PPO, and our proposed MM-STMAP model. The evaluation includes both total queue length distributions and average queue length trends over time. These results are presented in Fig. 6. As shown in the figure, MM-STMAP consistently outperforms all baseline methods under both experimental scenarios. In the box plots, MM-STMAP achieves the lowest total queue lengths, with narrower distributions and significantly reduced variance, indicating better control stability and efficiency. In the line plots, MM-STMAP maintains a steadily decreasing average queue length throughout the simulation. In contrast, other methods–particularly DQN and PPO–exhibit more fluctuation and higher congestion in

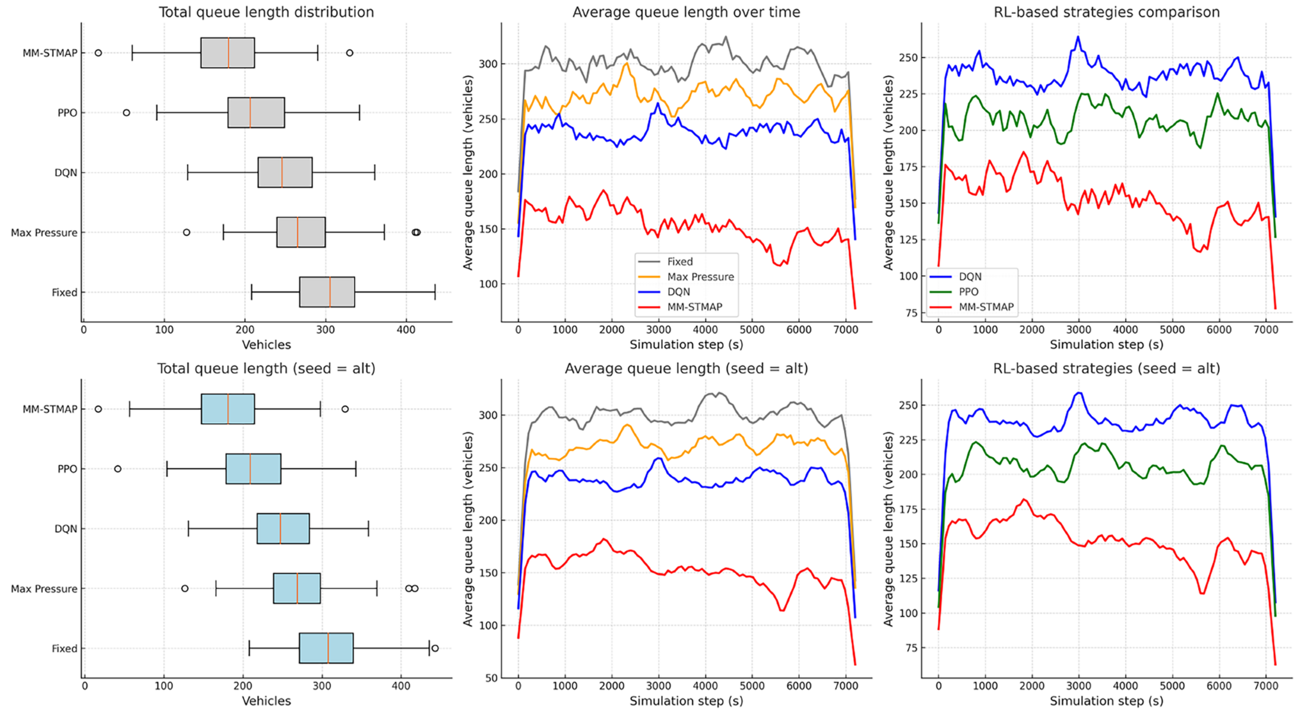


Fig. 6. Comparison of total and average queue lengths under different signal control strategies across two random seeds.

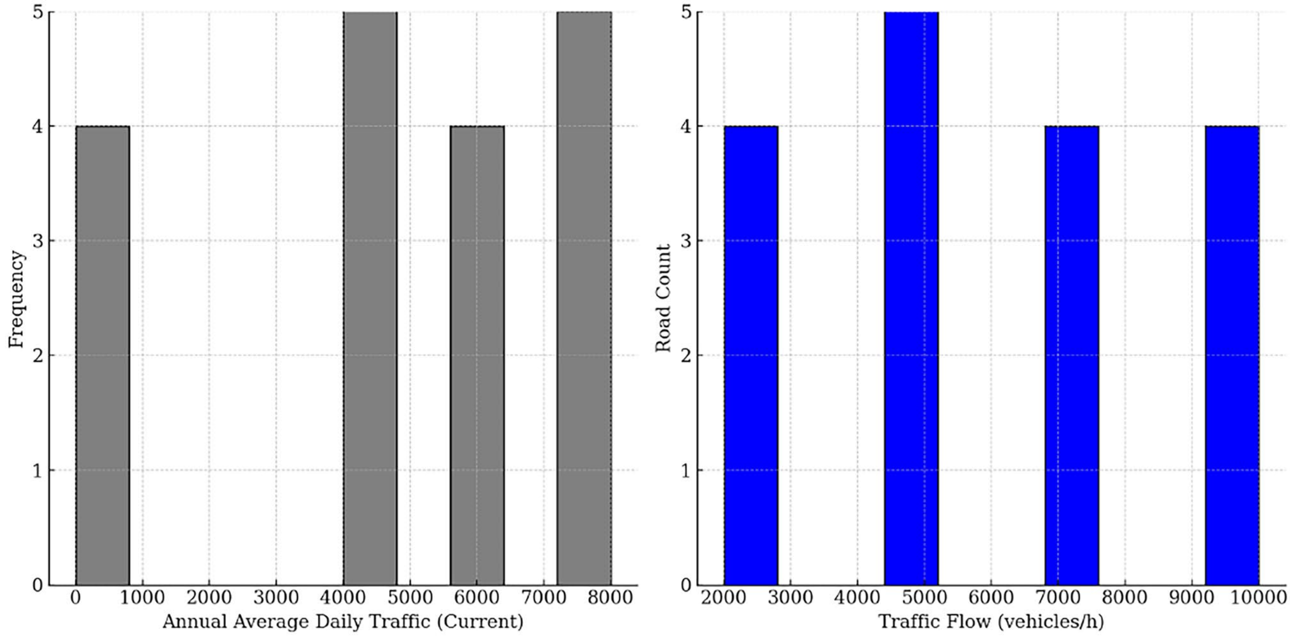
Variant	PeMSD4			METR-LA		
	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
MM-STMAP (Full)	18.0±0.2*	29.1±0.3*	10.2±0.1*	20.5±0.2*	32.4±0.3*	12.0±0.1*
No Spatio-Temporal GCN Module	20.2±0.3	31.5±0.4	12.4±0.2	22.3±0.3	34.1±0.4	13.5±0.2
No Linear Attention Mechanism	21.3±0.3	32.1±0.4	13.3±0.2	23.6±0.3	35.2±0.4	14.3±0.2
No MAPPO (Multi-Agent Optimization)	22.5±0.4	33.2±0.5	13.8±0.2	24.0±0.4	36.0±0.5	14.7±0.2
Single-Agent PPO (Replace MAPPO)	23.1±0.4	34.0±0.5	14.5±0.2	24.8±0.4	36.8±0.5	15.2±0.2

Table 7. Ablation Experiment Results for PeMSD4 and METR-LA Datasets. Values after ± represent 95% confidence intervals; * indicates $p < 0.05$ (statistically significant compared to all variants)

the later simulation stages. Furthermore, the consistency of MM-STMAP's performance across two distinct experimental settings highlights its strong robustness and generalization capability. These results demonstrate that the integration of multimodal spatiotemporal perception and multi-agent reinforcement learning enables MM-STMAP to achieve both high traffic throughput and adaptive stability. The model thus presents a scalable and practical solution for real-world urban traffic signal optimization.

Ablation studies and model components analysis

Based on the ablation experiment results shown in Table7, we can observe the performance of different variants of the MM-STMAP model (with specific modules removed or replaced) on the PeMSD4 and METR-LA datasets, with three key metrics (MAE, RMSE, MAPE) evaluated alongside 95% confidence intervals (±0.1–0.5) to ensure the reliability of performance differences. Specifically, MM-STMAP (Full) achieves the optimal performance on both datasets, with the lowest values in all metrics, and these advantages are statistically significant ($p < 0.05$), marked by *. Firstly, removing the Spatio-Temporal GCN module leads to a notable performance degradation. On PeMSD4, MAE increases by 12.2% (from 18.0±0.2 to 20.2±0.3) and RMSE rises by 8.3% (from 29.1±0.3 to 31.5±0.4); on METR-LA, MAE and RMSE increase by 8.8% and 5.2%, respectively. This confirms that the Spatio-Temporal GCN module is critical for capturing complex spatiotemporal dependencies in traffic networks, as it effectively models the spatial correlations between adjacent road segments and the temporal evolution of traffic flow. Secondly, the removal of the Linear Attention Mechanism further deteriorates model performance. Compared to the variant without Spatio-Temporal GCN, MAE increases by 5.4% (PeMSD4) and 5.8% (METR-LA), while RMSE rises by 1.9% and 3.2% respectively. This indicates that the Linear Attention Mechanism not only reduces the computational complexity of long-time-series processing (from quadratic to linear) but also enhances the model's ability to capture long-term temporal dependencies and adapt to external factors (e.g.,



(a) (b)

Fig. 7. Comparison of traffic flow distribution before and after optimization. The left subplot shows the distribution of Annual Average Daily Traffic, while the right subplot illustrates the optimized traffic flow. The optimization measures lead to a more balanced traffic flow and improved overall traffic efficiency.

weather, holidays), complementing the spatiotemporal modeling capability of the GCN module. Thirdly, the impact of multi-agent optimization is verified by two variants: removing MAPPO and replacing it with a single-agent PPO. Compared to the full model, the variant without MAPPO shows a 25.0% increase in MAE (PeMSD4) and 17.1% increase (METR-LA). More importantly, the single-agent PPO variant performs even worse than the MAPPO-removed version: MAE is further increased by 2.7% (PeMSD4) and 3.3% (METR-LA), and MAPE rises by 5.1% and 3.4% respectively. This demonstrates that multi-agent coordination (via MAPPO) is more effective than single-agent control in handling distributed urban traffic networks–single-agent models oversimplify network dynamics by treating the entire system as a single decision unit, failing to capture local-global interdependencies between intersections, while MAPPO enables adaptive coordination among multiple signal agents, thereby optimizing overall traffic efficiency. Finally, MM-STMAP (Full) maintains the best performance across all metrics, confirming that the Spatio-Temporal GCN module, Linear Attention Mechanism, and MAPPO multi-agent framework are mutually complementary and synergistic. The ablation results validate the rationality of each module's design and the necessity of integrating them into a unified framework for traffic flow prediction and signal control optimization.

Based on the data comparison in the Fig. 7, we present the changes in traffic flow distribution before and after optimization. The left subplot shows the distribution of Annual Average Daily Traffic (AADT), while the right subplot illustrates the change in Traffic Flow (vehicles/hour). By comparing these two plots, we can clearly observe the differences before and after optimization. In the pre-optimization scenario (left subplot), the traffic flow distribution is more scattered, with most of the traffic concentrated in lower value ranges. However, there are still some road sections with higher traffic flow, indicating an imbalance in the traffic flow. In contrast, the post-optimization scenario (right subplot) shows a more even distribution of traffic flow. After optimization, more road sections exhibit improved traffic flow, particularly in areas with higher traffic volumes, showing that the optimization measures significantly increased traffic flow and reduced congestion. This optimization has improved overall traffic flow efficiency. Especially in high-traffic areas during peak hours, the optimized traffic flow has increased the overall throughput, while reducing vehicle stops. The optimized traffic flow distribution shows a lower concentration of traffic, indicating that the optimization measures effectively balanced the traffic flow and reduced excessive congestion on certain road sections. These results demonstrate that the optimized traffic management solutions can improve traffic flow across large areas, enhance overall traffic efficiency, and reduce congestion, providing a more effective solution for urban traffic management.

Discussion and limitations

The MM-STMAP model has demonstrated promising performance in traffic flow prediction and signal control optimization tasks in theoretical and experimental validations, particularly after integrating multimodal information, spatiotemporal graph convolutional networks, and deep reinforcement learning techniques. Its core strengths lie in the integrated framework of spatiotemporal prediction and multi-agent control, which enables bidirectional synergy between traffic state perception and decision-making; the weighted linear attention mechanism that achieves efficient processing of long time-series data and adaptation to multi-modal information; and the multi-agent coordination mechanism embedded with spatiotemporal constraints that balances local and network-wide traffic dynamics.

However, despite its promising performance in theoretical and experimental validation, several limitations remain. First, the computational complexity of the MM-STMAP model is relatively high, especially when applied to large-scale urban traffic networks. As the size of the traffic network and the time steps increase, the computational resources required for model training and inference also increase significantly, potentially becoming a bottleneck in practical applications. Moreover, the MAPPO-based centralized training architecture may face communication bottlenecks when scaling to city-level networks with hundreds of intersections. Specifically, high-frequency transmission of local observation data and experience tuples from numerous agents to the central server, as well as synchronous policy parameter broadcasting, may lead to latency and bandwidth pressure in real-world infrastructure setups. Second, the model is highly dependent on data quality; in practical scenarios, traffic data often contains missing values and noise, and the timeliness and completeness of multimodal data sources may further affect performance, while the current data preprocessing strategies are insufficient to handle complex real-world data issues. It is worth investigating more robust data enhancement and repair methods tailored to traffic scenarios in future research. Third, the model's generalizability across diverse urban scenarios remains untested–current validation is based on datasets from specific regions (e.g., LA), and its adaptability to cities with distinct traffic characteristics (such as different road network structures, peak-hour patterns, or regional management policies) has not been fully verified. Exploring transfer learning or domain adaptation approaches to enhance cross-city applicability is a valuable direction for future exploration. Fourth, the model's ability to handle sudden traffic changes and extreme conditions (e.g., traffic accidents, large-scale events) needs further exploration, as its response time and adaptability to unexpected events have not been thoroughly evaluated. Future studies could integrate real-time emergency information and design adaptive decision mechanisms to improve the model's resilience. Additionally, while the model indirectly achieves emission reduction through optimizing traffic efficiency, future research could explicitly incorporate emission-related metrics into the optimization objective and validate with real-world environmental monitoring data to further quantify its ecological benefits.

To address the aforementioned limitations and explore the practical application potential of the model, future work will focus on the following directions: First, improve computational efficiency by adopting more efficient training algorithms, model compression, or distributed computing techniques to enhance applicability in large-scale transportation networks. Second, mitigate communication bottlenecks in centralized training by introducing hierarchical agent clustering (grouping geographically adjacent intersections into sub-networks) and asynchronous communication protocols, reducing the frequency of global data synchronization while maintaining coordination effectiveness. Third, optimize data transmission by adopting lightweight feature representation and incremental update strategies, minimizing the amount of data exchanged between agents and the central server. Fourth, conduct systematic empirical analysis on the model's sensitivity to data quality, and develop targeted mitigation strategies–including graph-based multi-scale data repair modules to fill missing traffic records using spatial correlations, and adaptive denoising mechanisms to suppress outlier interference. Fifth, extend experimental validation to multiple representative datasets from different cities, and explore transfer learning and domain adaptation techniques to enhance cross-domain robustness. Sixth, strengthen the model's emergency response capability by integrating real-time external information (e.g., traffic accident reports, vehicle location data) and developing flexible adaptive mechanisms to handle sudden traffic changes and extreme conditions. Seventh, carry out practical deployment tests in real urban traffic scenarios to verify the model's actual application effect and optimize it according to on-site feedback.

Through these improvements, the MM-STMAP model's applicability and performance are expected to be further enhanced, laying a solid foundation for its potential practical application in urban traffic management and intelligent transportation systems.

Conclusion

This study proposes a city traffic management optimization model (MM-STMAP) based on multimodal learning and deep reinforcement learning, which effectively captures the spatiotemporal dependencies of traffic flow through spatiotemporal graph convolutional networks and weighted linear attention mechanisms. This model adopts multi-agent reinforcement learning to achieve coordinated signal control optimization. The experimental results show that MM-STMAP outperforms existing methods in traffic flow prediction and signal control tasks, demonstrating excellent adaptability and robustness in complex dynamic environments. By reducing vehicle idle time and frequent stops, the model also realizes indirect ecological benefits, providing empirical support for the sustainability claims highlighted in the abstract. However, there are still challenges such as data quality dependency, computational complexity, and potential communication bottlenecks in large-scale deployment. Future research will focus on optimizing model architecture, exploring cross-modal learning and interpretability techniques, and addressing the aforementioned limitations to promote its practical applications in intelligent transportation systems. Additionally, further investigations into integrating explicit sustainability objectives and enhancing cross-scenario generalizability will help expand the model's practical value and research impact.

Data availability

The datasets used in this study are publicly available. The PeMSD4 dataset is available from the California Performance Measurement System (PeMS) at https://pems.dot.ca.gov/. The METR-LA dataset is available from the Los Angeles Traffic Monitoring System at https://github.com/liyaguang/DCRNN/tree/master/data.

Received: 18 December 2025; Accepted: 23 January 2026
Published online: 06 February 2026

References

1. Wu, K. et al. Big-data empowered traffic signal control could reduce urban carbon emission. *Nat. Commun.* **16**, 2013 (2025).
2. Wang, X. et al. Traffic light optimization with low penetration rate vehicle trajectory data. *Nat. Commun.* **15**, 1306 (2024).
3. Li, H., Chen, Y., Li, K., Wang, C. & Chen, B. Transportation internet: A sustainable solution for intelligent transportation systems. *IEEE Trans. Intell. Transp. Syst.* **24**, 15818–15829 (2023).
4. Wang, F.-Y. *et al.* Transportation 5.0: The dao to safe, secure, and sustainable intelligent transportation systems. *IEEE Transactions on Intelligent Transportation Systems***24**, 10262–10278 (2023).
5. Wang, X. et al. Parallel driving with big models and foundation intelligence in cyber-physical-social spaces. *Research* **7**, 0349 (2024).
6. Selvan, C. et al. Traffic prediction using gps based cloud data through rnn-lstm-cnn models: Addressing road congestion, safety, and sustainability in smart cities. *SN Computer Sci.* **6**, 159 (2025).
7. Bahe, M. *et al.* Traffic prediction of real network traffic data with lstm neural networks. In *Future of Information and Communication Conference*, 311–326 (Springer, 2024).
8. Ma, C., Gu, K., Zhao, Y. & Wang, T. Research on highway traffic flow prediction based on a hybrid model of arima–gwo–lstm. *Neural Computing and Applications* 1–20 (2024).
9. Huang, H., Chen, J., Sun, R. & Wang, S. Short-term traffic prediction based on time series decomposition. *Physica A* **585**, 126441 (2022).
10. Kishore Anthuvan Sahayaraj, K., Chodnekar, A. & Mishra, A. Optimizing urban traffic flow prediction: Integrating spatial–temporal analysis with a hybrid gnn and gated-attention gru model. In *International Conference on Smart Data Intelligence*, 381–391 (Springer, 2024).
11. Karim, A. A. & Nower, N. Robust traffic prediction using probabilistic spatio-temporal graph convolutional network. In *International Conference on Engineering Applications of Neural Networks*, 259–273 (Springer, 2024).
12. Zheng, H., Li, X., Li, Y., Yan, Z. & Li, T. Gcn-gan: integrating graph convolutional network and generative adversarial network for traffic flow prediction. *IEEE Access* **10**, 94051–94062 (2022).
13. Zhao, J., Yu, Z., Yang, X., Gao, Z. & Liu, W. Short term traffic flow prediction of expressway service area based on stl-oms. *Physica A* **595**, 126937 (2022).
14. Liu, H., Zhang, X.-Y., Yang, Y.-X., Li, Y.-F. & Yu, C.-Q. Hourly traffic flow forecasting using a new hybrid modelling method. *J. Central South Univ.* **29**, 1389–1402 (2022).
15. Luo, L., Han, S., Li, Z., Yang, J. & Yang, X. A traffic flow prediction framework based on clustering and heterogeneous graph neural networks. In *International Conference on Intelligent Computing*, 58–69 (Springer, 2023).
16. Ning, X., Gao, S., Liu, J., Cheng, L. & Zhang, Y. Few-shot agricultural disease detection method using contextual attention generation. *Alex. Eng. J.* **130**, 101–114 (2025).
17. Costa, D. G. et al. A survey of emergencies management systems in smart cities. *IEEE Access* **10**, 61843–61872 (2022).
18. Li, M., Chen, Q., Li, G. & Han, D. Umformer: a transformer dedicated to univariate multistep prediction. *IEEE Access* **10**, 101347–101361 (2022).
19. Fadhel, M. A. *et al.* Comprehensive systematic review of information fusion methods in smart cities and urban environments. *Information Fusion* 102317 (2024).
20. Yoon, C. *et al.* V-dcrnn: Virtual network-based diffusion convolutional recurrent neural network for estimating unobserved traffic data. *IEEE Transactions on Intelligent Transportation Systems* (2025).
21. Xu, Q., Pang, Y., Zhou, X. & Liu, Y. Pigat: Physics-informed graph attention transformer for air traffic state prediction. *IEEE Trans. Intell. Transp. Syst.* **25**, 12561–12577 (2024).
22. Cai, C. et al. Ta-astgcn: a trend-aware adaptive spatio-temporal graph convolutional network for traffic flow prediction. *Neural Comput. Appl.* **37**, 23335–23358 (2025).
23. Gao, X. *et al.* A survey of collaborative perception in intelligent vehicles at intersections. *IEEE Transactions on Intelligent Vehicles* (2024).
24. Zhang, Y., Jiang, C., Yue, B., Wan, J. & Guizani, M. Information fusion for edge intelligence: A survey. *Information Fusion* **81**, 171–186 (2022).
25. Pan, Y. A., Li, F., Li, A., Niu, Z. & Liu, Z. Urban intersection traffic flow prediction: A physics-guided stepwise framework utilizing spatio-temporal graph neural network algorithms. *Multimodal Transportation* **4**, 100207 (2025).
26. Ning, E., Miao, J., Xie, S., Ma, H. & Ning, X. Occluded person re-identification in multi-scenarios: A synergistic interaction framework with perception-aware optimization. *Eng. Appl. Artif. Intell.* **166**, 113674 (2026).
27. Liu, S. & Wang, X. An improved transformer based traffic flow prediction model. *Sci. Rep.* **15**, 8284 (2025).
28. Liu, T. & Meidani, H. Multi-class traffic assignment using multi-view heterogeneous graph attention networks. *Expert Syst. Appl.* **286**, 128072 (2025).
29. Zhang, Z. & Luo, F. Multisource data-driven visibility prediction: integrating channel attention mechanism with multiscale temporal convolution. In *Fourth International Conference on Advanced Algorithms and Neural Networks (AANN 2024)*, 13416, 338–344 (SPIE, 2024).
30. Pan, Y. A. et al. A fundamental diagram based hybrid framework for traffic flow estimation and prediction by combining a markovian model with deep learning. *Expert Syst. Appl.* **238**, 122219 (2024).
31. Liu, T. & Meidani, H. End-to-end heterogeneous graph neural networks for traffic assignment. *Transp. Res. Part C: Emerg. Technol.* **165**, 104695 (2024).
32. Javaheri, A. *et al.* Evaluating impact of smart parking systems on parking violations. *IEEE Access* (2024).
33. Xu, Q., Pang, Y. & Liu, Y. Air traffic density prediction using bayesian ensemble graph attention network (began). *Transp. Res. Part C: Emerg. Technol.* **153**, 104225 (2023).
34. Huang, Y., Dong, Y., Tang, Y. & Li, L. Leverage multi-source traffic demand data fusion with transformer model for urban parking prediction. *arXiv preprint* arXiv:2405.01055 (2024).
35. Xie, Z., Ma, Y., Zhang, Z. & Chen, S. Real-time driving risk prediction using a self-attention-based bidirectional long short-term memory network based on multi-source data. *Accident Analysis & Prevention* **204**, 107647 (2024).
36. Lv, S., Wang, K., Yang, H. & Wang, P. An origin-destination passenger flow prediction system based on convolutional neural network and passenger source-based attention mechanism. *Expert Syst. Appl.* **238**, 121989 (2024).
37. Gan, S. et al. Multisource adaption for driver attention prediction in arbitrary driving scenes. *IEEE Trans. Intell. Transp. Syst.* **23**, 20912–20925 (2022).
38. Li, H., Ge, Y., Duan, Y. & Zhang, X. Collaborative optimization of signals and ecological driving speed guidance for buses without dedicated bus lanes in the connected environment. *Sci. Rep.* **14**, 30995 (2024).
39. Chauhan, S. S. & Kumar, D. Mode search optimization algorithm for traffic prediction and signal controlling using bellman-ford with tpfn path discovery model based on deep lstm classifier. *SN Computer Science* **4**, 686 (2023).
40. Tomar, I., Indu, S. & Pandey, N. Traffic signal control methods: Current status, challenges, and emerging trends. *Proceedings of Data Analytics and Management: ICDAM 2021, Volume 1* 151–163 (2022).

41. Bajada, J., Grech, J. & Bajada, T. Deep reinforcement learning of autonomous control actions to improve bus-service regularity. In *European Conference on Artificial Intelligence*, 138–155 (Springer, 2023).
42. Li, S., Hu, J., Zhang, B., Ning, X. & Wu, L. Dynamic personalized federated learning for cross-spectral palmprint recognition. *IEEE Transactions on Image Processing* (2025).
43. Ghadbane, H. E. et al. Energy management of electric vehicle using a new strategy based on slap swarm optimization and differential flatness control. *Sci. Rep.* **14**, 3629 (2024).
44. Noaeen, M. et al. Reinforcement learning in urban network traffic signal control: A systematic literature review. *Expert Syst. Appl.* **199**, 116830 (2022).
45. Cai, C. & Wei, M. Adaptive urban traffic signal control based on enhanced deep reinforcement learning. *Sci. Rep.* **14**, 14116 (2024).
46. Bouktif, S., Cheniki, A., Ouni, A. & El-Sayed, H. Deep reinforcement learning for traffic signal control with consistent state and reward design approach. *Knowl.-Based Syst.* **267**, 110440 (2023).
47. Liu, T., Wang, Y., Zhou, H., Luo, J. & Deng, F. Distributed short-term traffic flow prediction based on integrating federated learning and tcn. *IEEE Access* (2024).
48. Yazdani, M. et al. Intelligent vehicle pedestrian light (ivpl): A deep reinforcement learning approach for traffic signal control. *Transportation research part C: emerging technologies* **149**, 103991 (2023).
49. Sattar, K. et al. Transparent deep machine learning framework for predicting traffic crash severity. *Neural Comput. Appl.* **35**, 1535–1547 (2023).
50. Mao, F., Li, Z. & Li, L. A comparison of deep reinforcement learning models for isolated traffic signal control. *IEEE Intell. Transp. Syst. Mag.* **15**, 160–180 (2022).
51. Du, X. et al. Felight: Fairness-aware traffic signal control via sample-efficient reinforcement learning. *IEEE Trans. Knowl. Data Eng.* **36**, 4678–4692 (2024).
52. Feng, J. et al. Evhf-gcn: an emergency vehicle priority scheduling model based on heterogeneous feature fusion with graph convolutional networks. *IEEE Access* **12**, 4166–4177 (2024).
53. Swapno, S. M. R. *et al.* Traffic light control using reinforcement learning. In *2024 international conference on integrated circuits and communication systems (ICICACS)*, 1–7 (IEEE, 2024).
54. Xu, Q., Pang, Y., Zhang, Z. & Liu, Y. Data-driven governing equation identification of near terminal air traffic flow dynamics. *J. Air Transp. Manag.* **129**, 102871 (2025).
55. Zhu, R. et al. Multi-agent broad reinforcement learning for intelligent traffic light control. *Inf. Sci.* **619**, 509–525 (2023).
56. Huang, Z., Wu, J. & Lv, C. Efficient deep reinforcement learning with imitative expert priors for autonomous driving. *IEEE Transactions on Neural Networks and Learning Systems* **34**, 7391–7403 (2022).
57. Li, A. *et al.* Matvt: A transformer-based approach for multi-agent prediction in complex traffic scenarios. *IEEE Transactions on Vehicular Technology* (2025).
58. Li, J., Yu, C., Shen, Z., Su, Z. & Ma, W. A survey on urban traffic control under mixed traffic environment with connected automated vehicles. *Transportation research part C: emerging technologies* **154**, 104258 (2023).
59. Cao, J., Zou, Q., Ning, X., Wang, Z. & Wang, G. Fipnet: Self-supervised low-light image enhancement combining feature and illumination priors. *Neurocomputing* **623**, 129426 (2025).
60. Shang, J. *et al.* Graph-based cooperation multi-agent reinforcement learning for intelligent traffic signal control. *IEEE Internet of Things Journal* (2025).
61. Zhang, X., Zhao, C., Liao, F., Li, X. & Du, Y. Online parking assignment in an environment of partially connected vehicles: A multi-agent deep reinforcement learning approach. *Transportation Research Part C: Emerging Technologies* **138**, 103624 (2022).
62. Qi, X., Hu, W., Li, B. & Han, K. Stgnn-fam: A traffic flow prediction model for spatiotemporal graph networks based on fusion of attention mechanisms. *J. Adv. Transp.* **2023**, 8880530 (2023).
63. Wang, X. et al. Dstspyn: a dynamic spatial-temporal similarity pyramid network for traffic flow prediction. *Appl. Intell.* **55**, 1–23 (2025).
64. Bernal-Sanchez, J. & Riascos-Ochoa, J. Traffic forecasting in bogota, colombia, with attention temporal graph convolutional networks (a3t-gcn). In *International Conference on Applied Informaticn* (2024).
65. Izadi, M., Safayani, M. & Mirzaei, A. Knowledge distillation on spatial-temporal graph convolutional network for traffic prediction. *Int. J. Comput. Appl.* **47**, 45–56 (2025).
66. Tan, C., Cao, Y., Ban, X. & Tang, K. Connected vehicle data-driven fixed-time traffic signal control considering cyclic time-dependent vehicle arrivals based on cumulative flow diagram. *IEEE Trans. Intell. Transp. Syst.* **25**, 8881–8897 (2024).
67. Genser, A. et al. A traffic signal and loop detector dataset of an urban intersection regulated by a fully actuated signal control system. *Data Brief* **48**, 109117 (2023).

Author contributions

Ruokun Wang contributed to the conceptualization, methodology, and model development. Ju Zhang and Xikui Wang were responsible for data collection, analysis, and model implementation. Haoyang He assisted with the literature review and system design, while Yuelin Zou provided research supervision and manuscript revision.

Funding

This work was supported by the Natural Science Foundation of the Jiangsu Higher Education Institution of China (Grant Number: 22KJB580008) the Scientific and technological innovation team project of Nanjing Vocational Institute of Railway Technology (Grant Number: CXTD2022003) the Natural Science Foundation of the Jiangsu Higher Education Institution of China (Grant Number: 23KJB580012) the Jiangsu Qinglan Project (Grant Number: 5002024012-RCQL01).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to R.W.